Introduction to Internet of Things

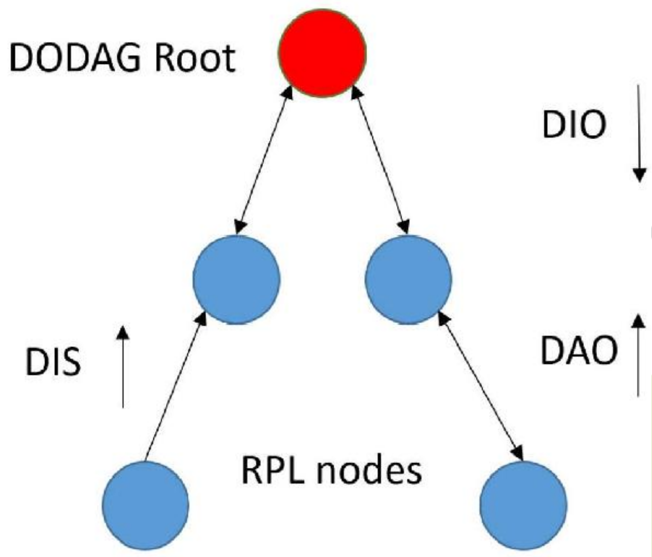
Lecture 11&12 (RPL)

RPL Protocol

- RPL is a routing protocol for Low Power and Lossy networks.
- Particularly designed for IoT networks where devices are:

1. Low power
2. Limited memory
3. Unreliable wireless links

- RPL is a distance vector routing protocol, and is beased on the tree like routing structure called DODAG (DIstance oriented Directed Acyclic Graphs)
- Directed Acyclic Graph means a graph with no cycle or no closed loop.



Basic terms in RPL

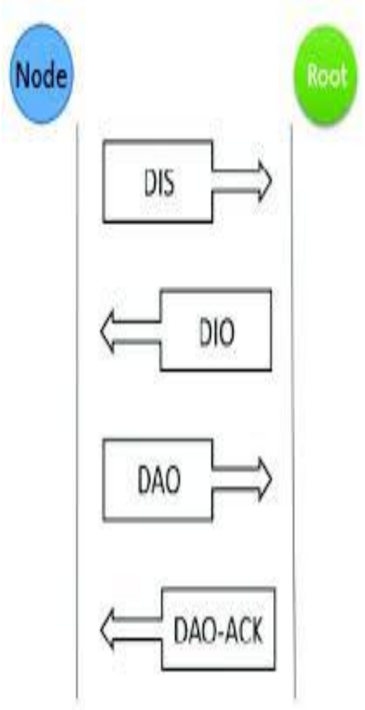
- Root: A node that is destination node of DAG. It has no outgoing edge.
- Up: It is any edge that is directed towards the root.
- Down: It is any edge that is directed away fron the root.
- Rank: It is the number of hopes from the root node.
- Grounded: When DODAG reaches its goal, it is called grounded.
- Floating DODAG: When DODAG is not connected or yet to reach its goal.

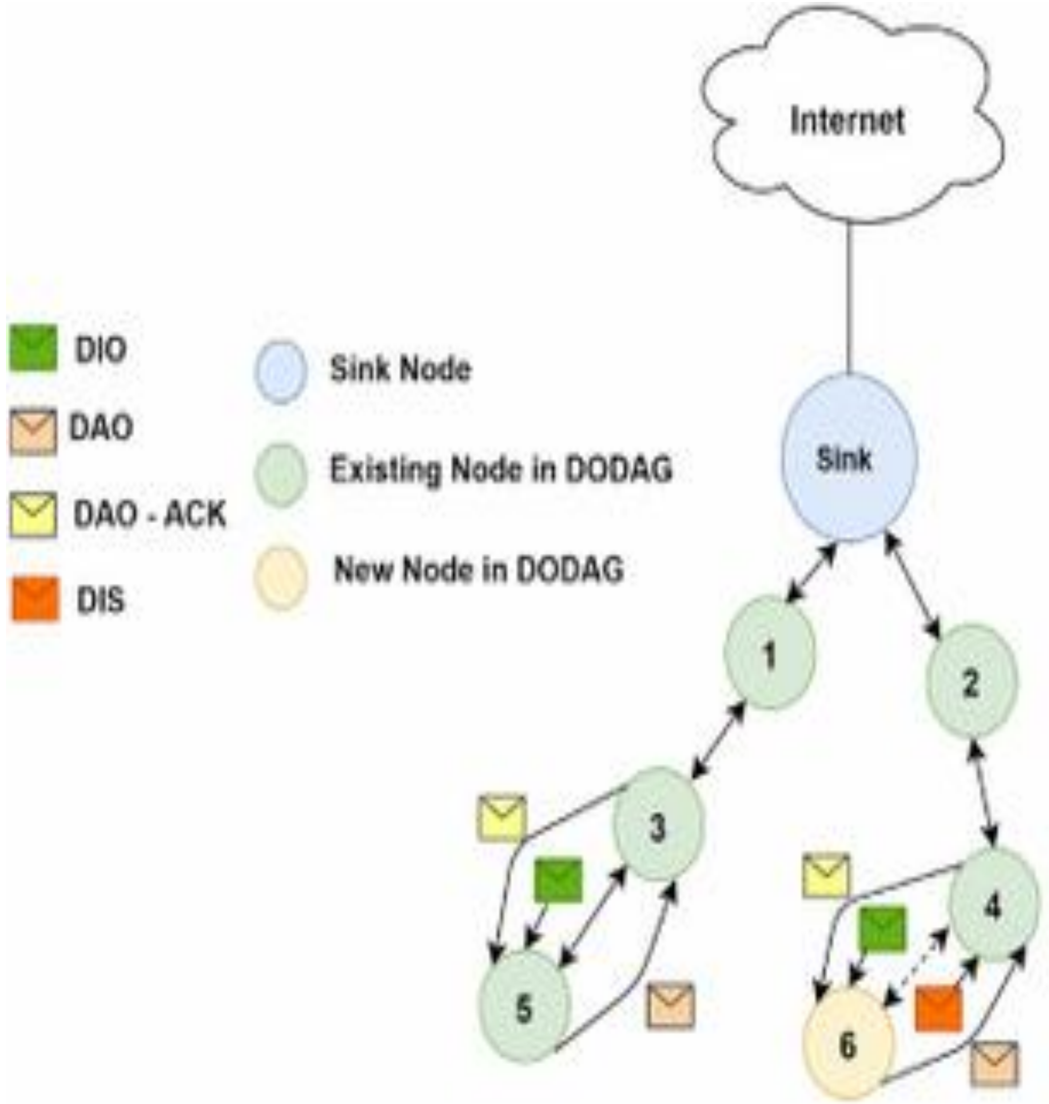
DODAG: A special kind of DAG where each node wants to reach a single destination.

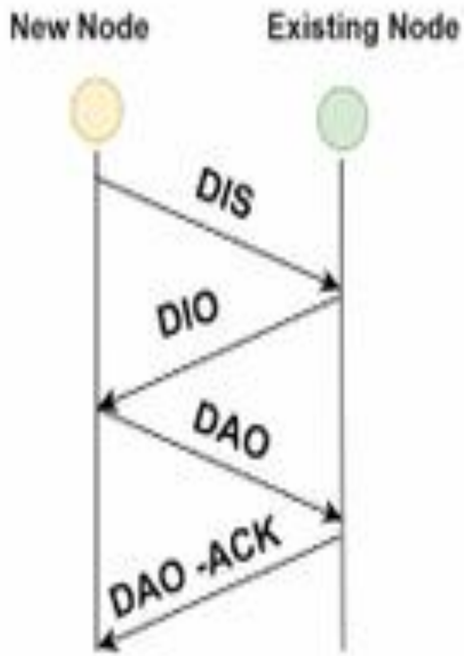
RPL instance: When we have one or more DODAGs, then each DODAG is an instance.

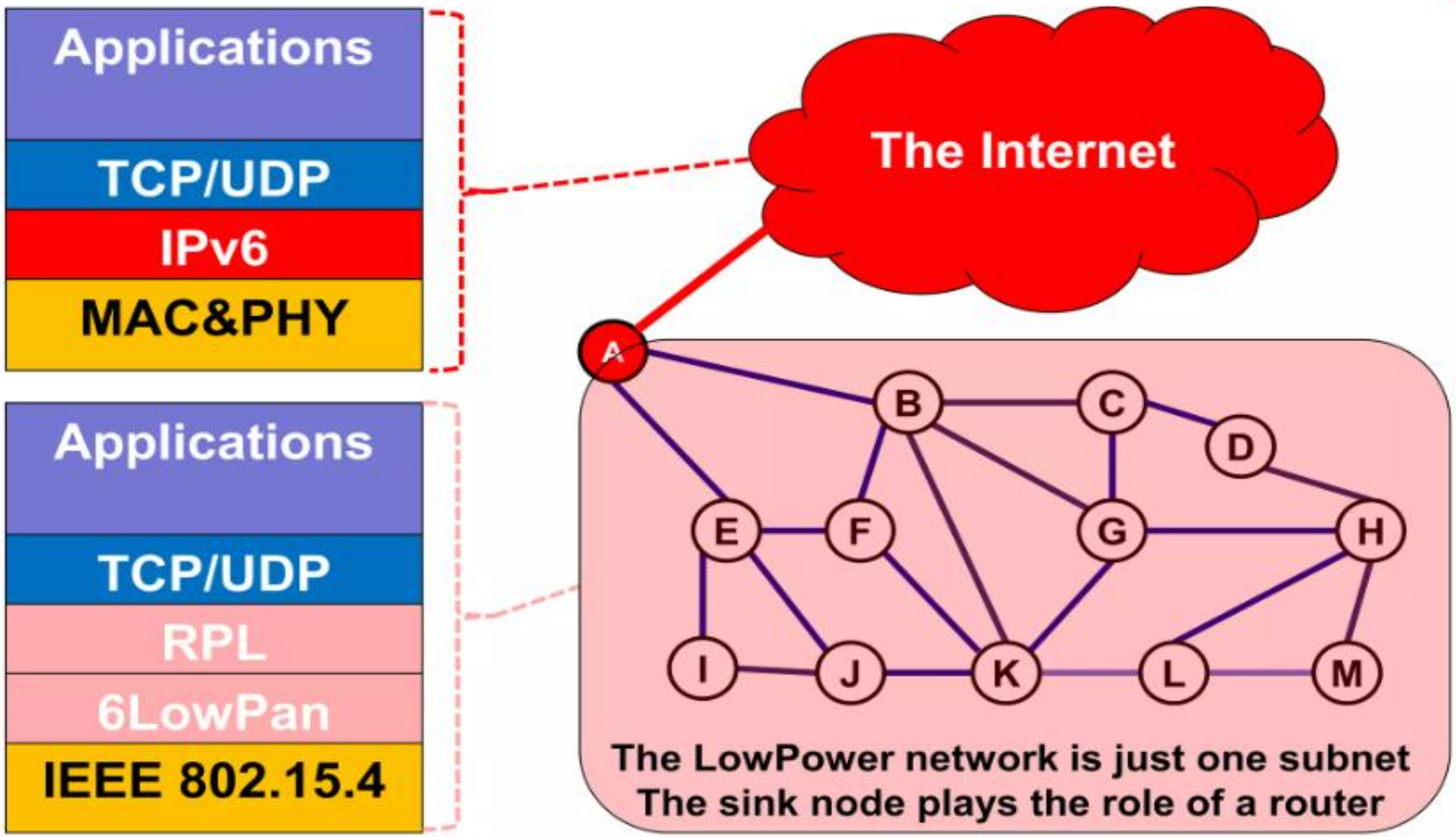
RPL control messages

- DIS (DODAG Information Solicitation): It is forwarded by new node to any older node. It wants to join DODAG.
- DIO (DODAG Information Object): This message can be forwarded by older node to new node as a reply. Also, it is used to make DODAG send by the root node to all other nodes.
- DODA Advertisement Object (DAO): It is a request send by the child node to the root node. This message requests to allow the child node to join to the DODAG.
- DAO-ACK: It is a response send by the root or parent node to a child ndoe.

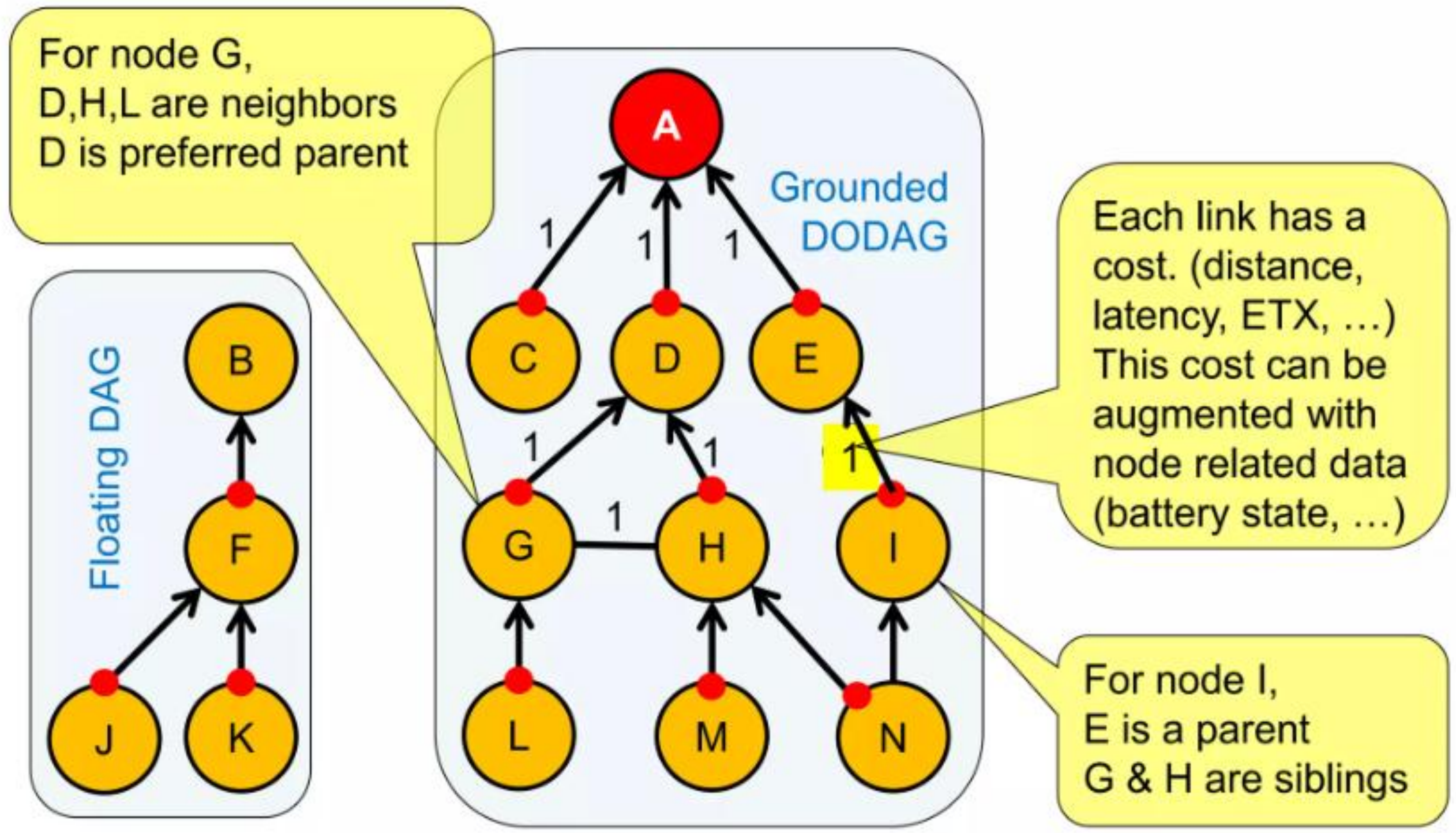


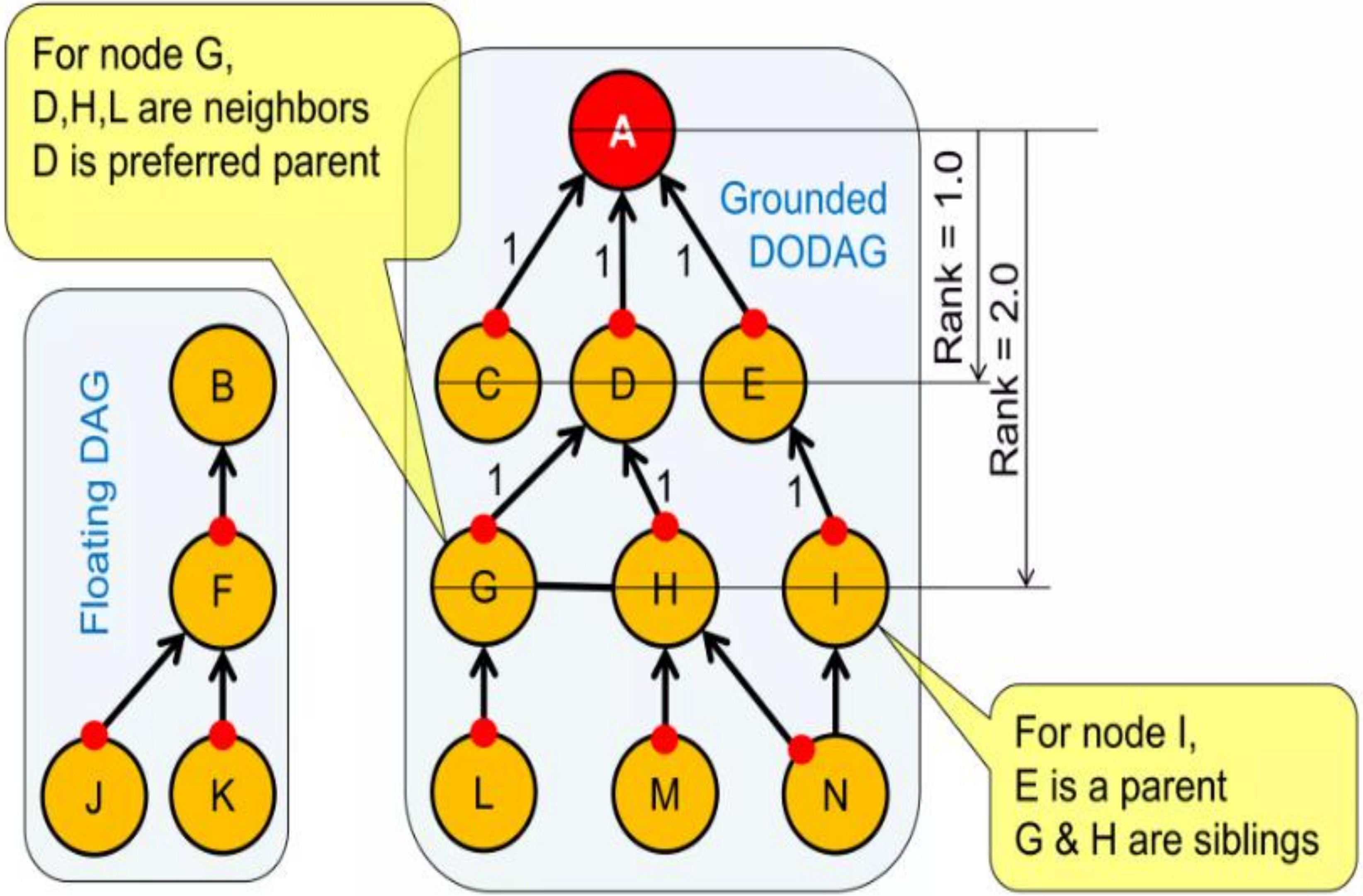

DIO
DAO
DAO - ACK
DIS
Sink Node
Existing Node in DODAG
New Node in DODAG
Internet
Sink
1
2
3
4
5
6


New Node
Existing Node
DIS
DIO
DAO
DAO -ACK

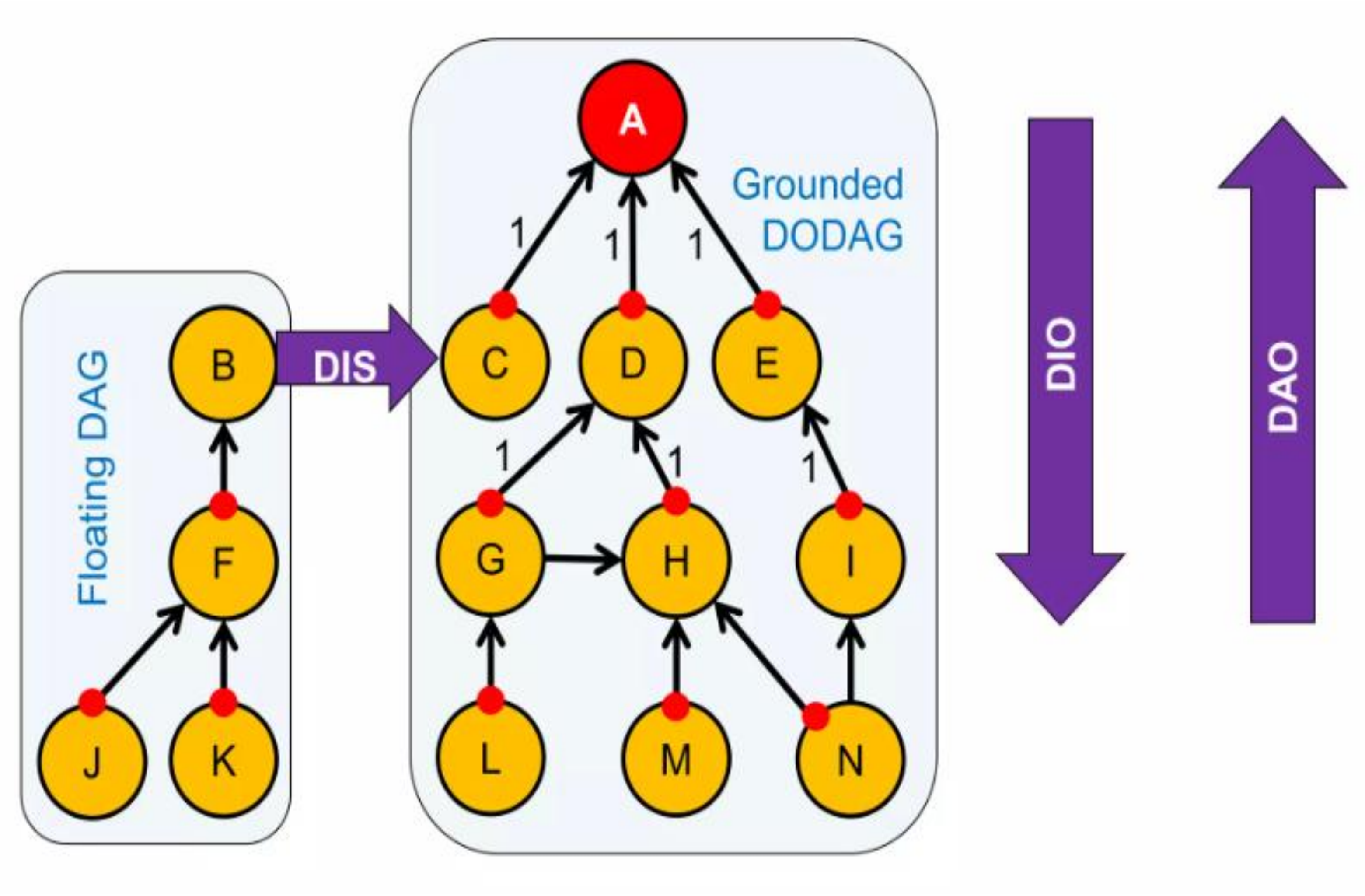


Applications
TCP/UDP
IPv6
MAC&PHY
Applications
TCP/UDP
RPL
6LowPan
IEEE 802.15.4
The Internet
A
B
C
D
E
F
G
H
I
J
K
L
M
The LowPower network is just one subnet
The sink node plays the role of a router

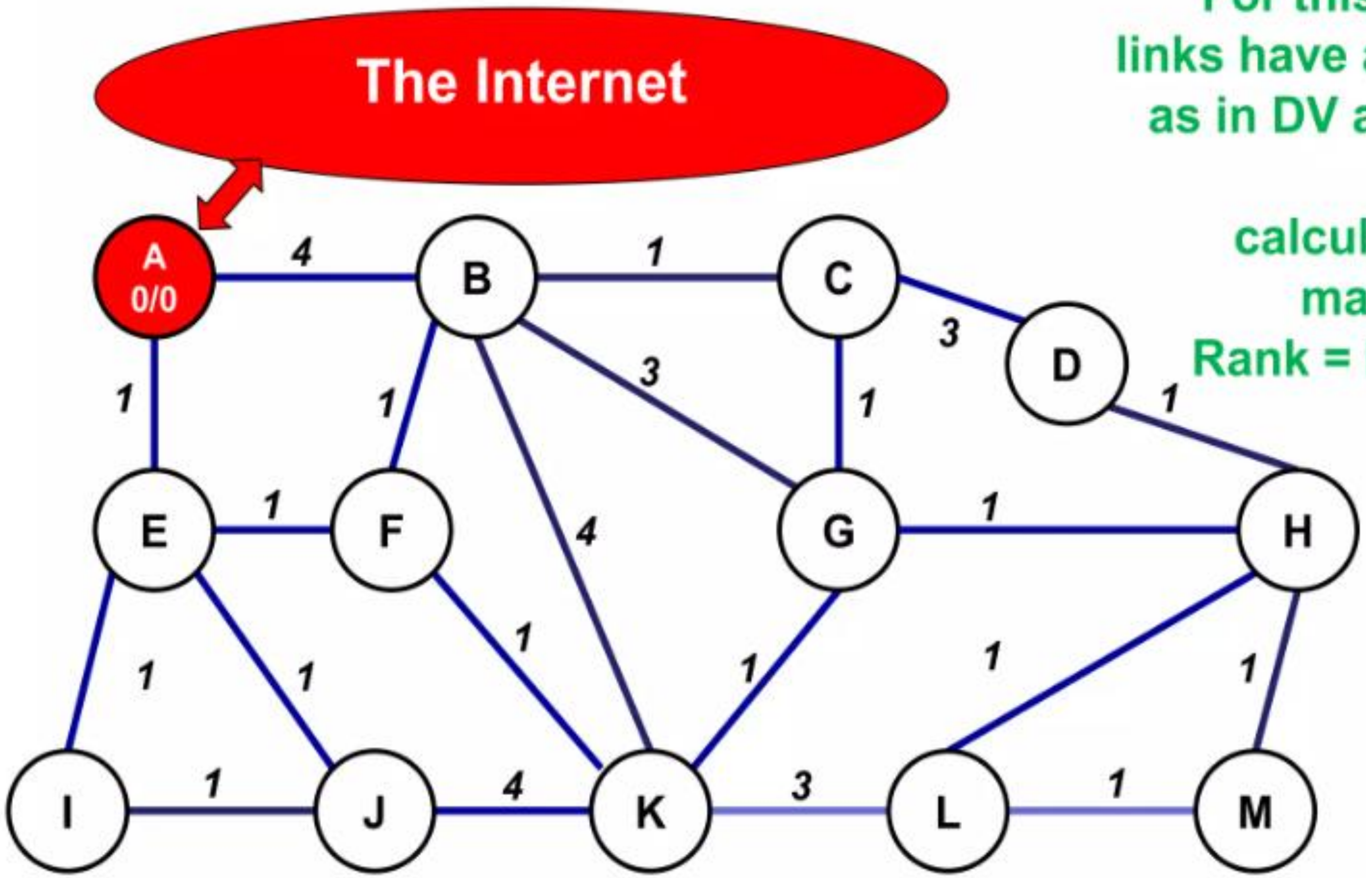

For node G,
D,H,L are neighbors
D is preferred parent
Grounded DODAG
Floating DAG
A
B
C
D
E
F
G
H
I
J
K
L
M
N
1
1
1
1
1
1
1
Each link has a cost. (distance, latency, ETX, …)
This cost can be augmented with node related data (battery state, …)
For node I,
E is a parent
G & H are siblings



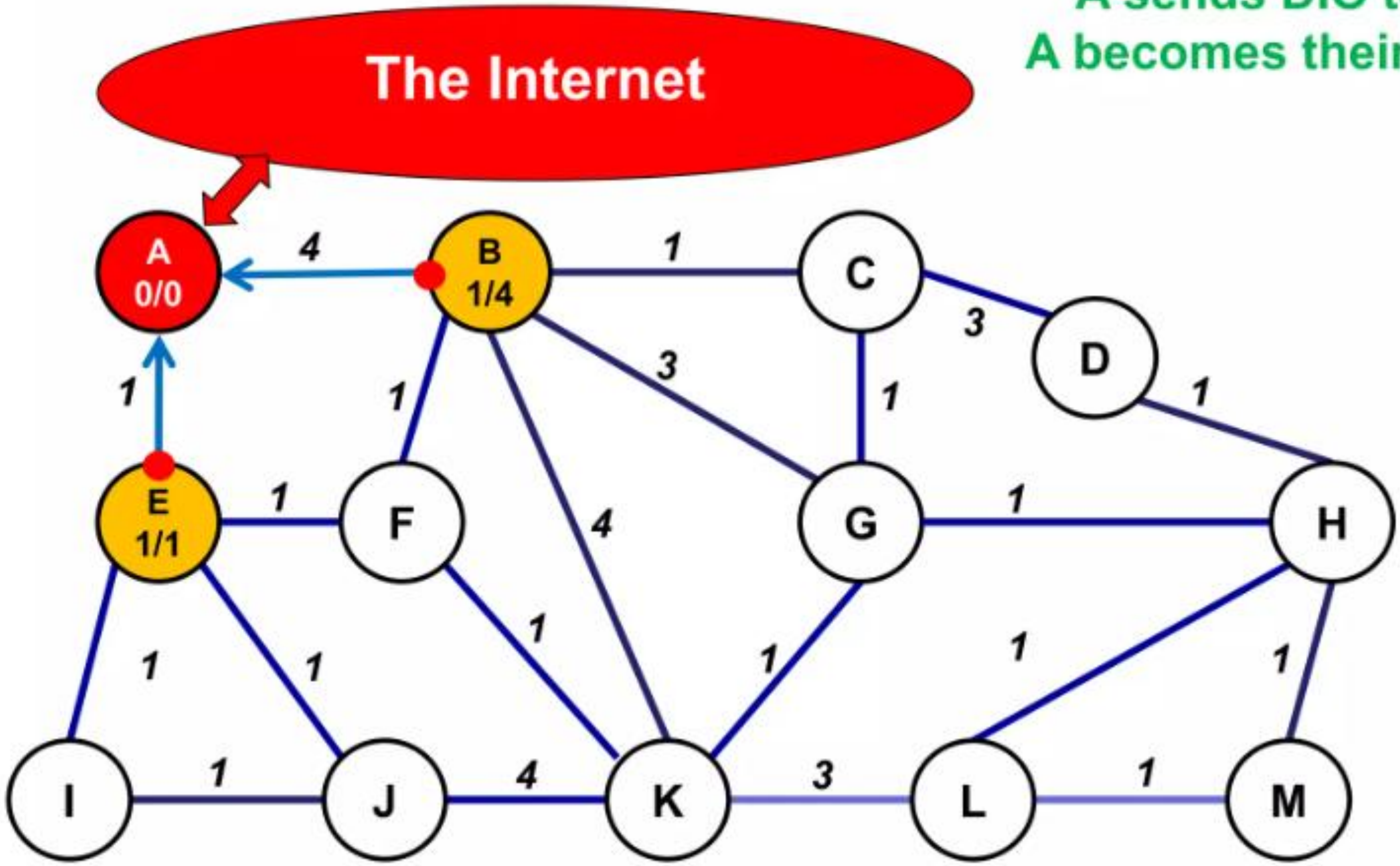
For node G,
D,H,L are neighbors
D is preferred parent
A
Grounded
DODAG
Rank = 1.0
Rank = 2.0
1
1
1
C
D
E
1
1
1
G
H
I
L
M
N
Floating DAG
B
F
J
K
For node I,
E is a parent
G & H are siblings


Floating DAG
B
F
J
K
DIS
Grounded
DODAG
A
C
D
E
G
H
I
L
M
N
1
1
1
1
1
1
DIO
DAO

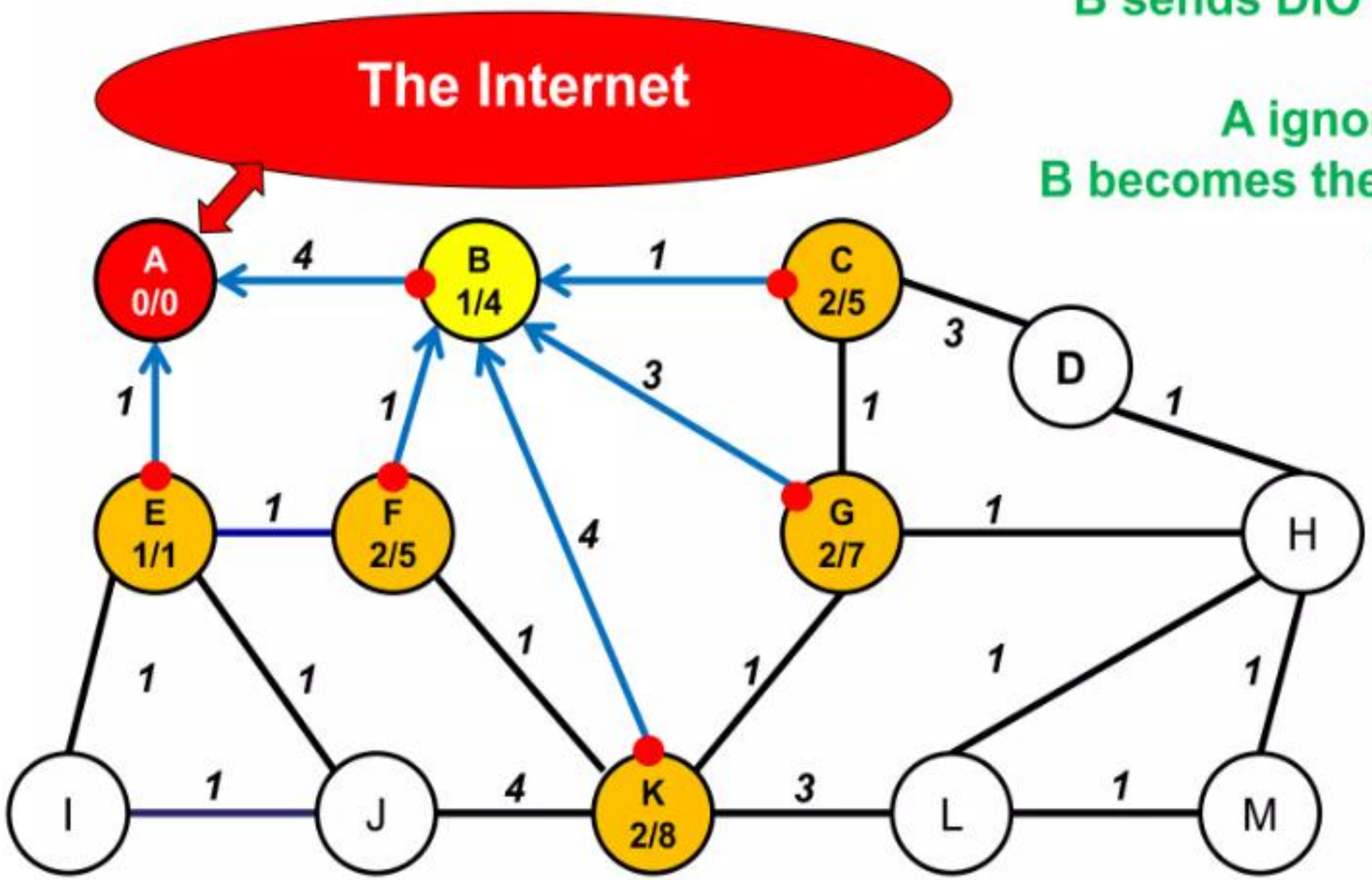
Example



For this example, links have a distance as in DV and AODV. The Rank calculations are made so that Rank = hop count

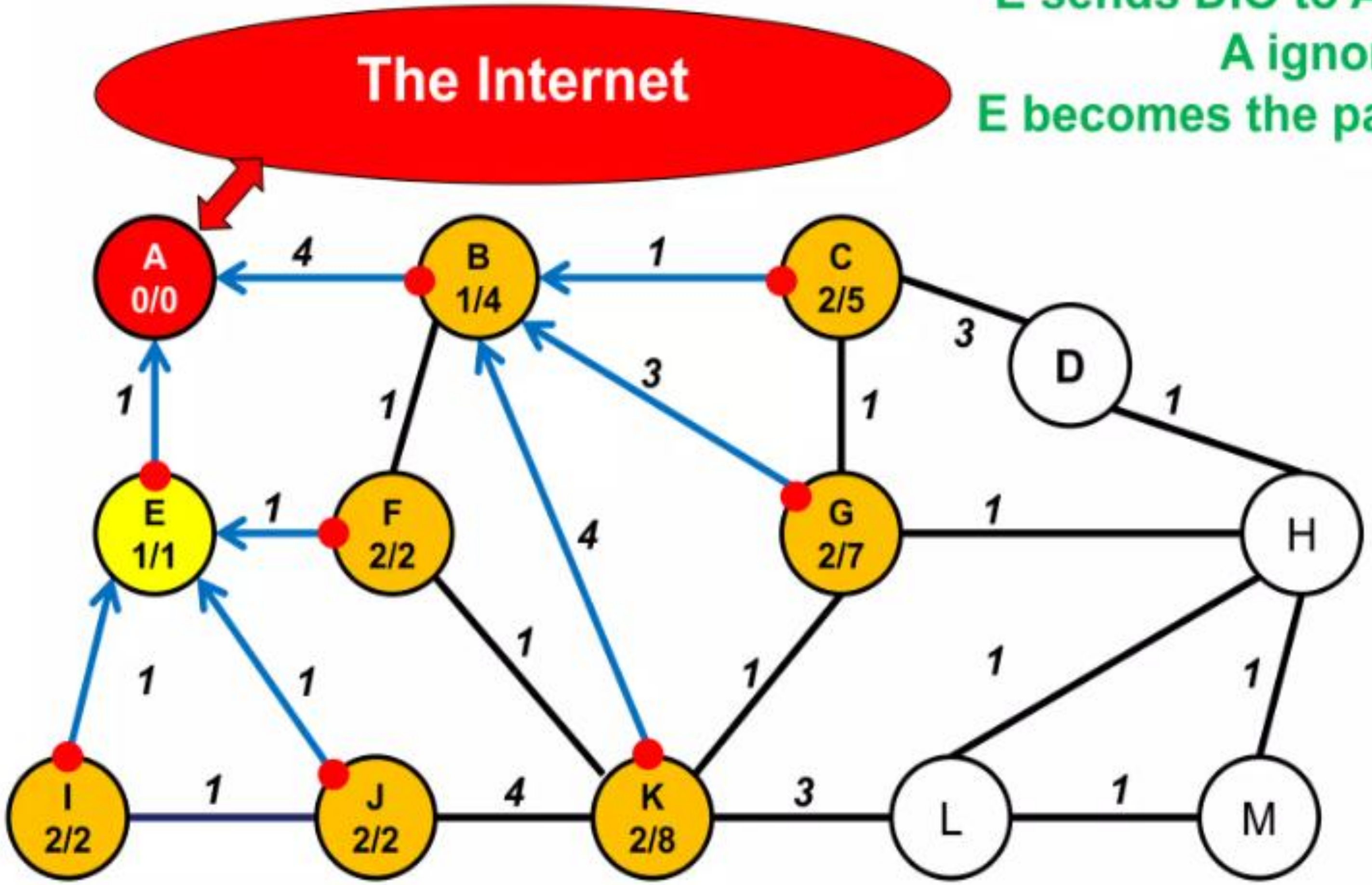


A sends DIO to B and E.
A becomes their preferred parent.

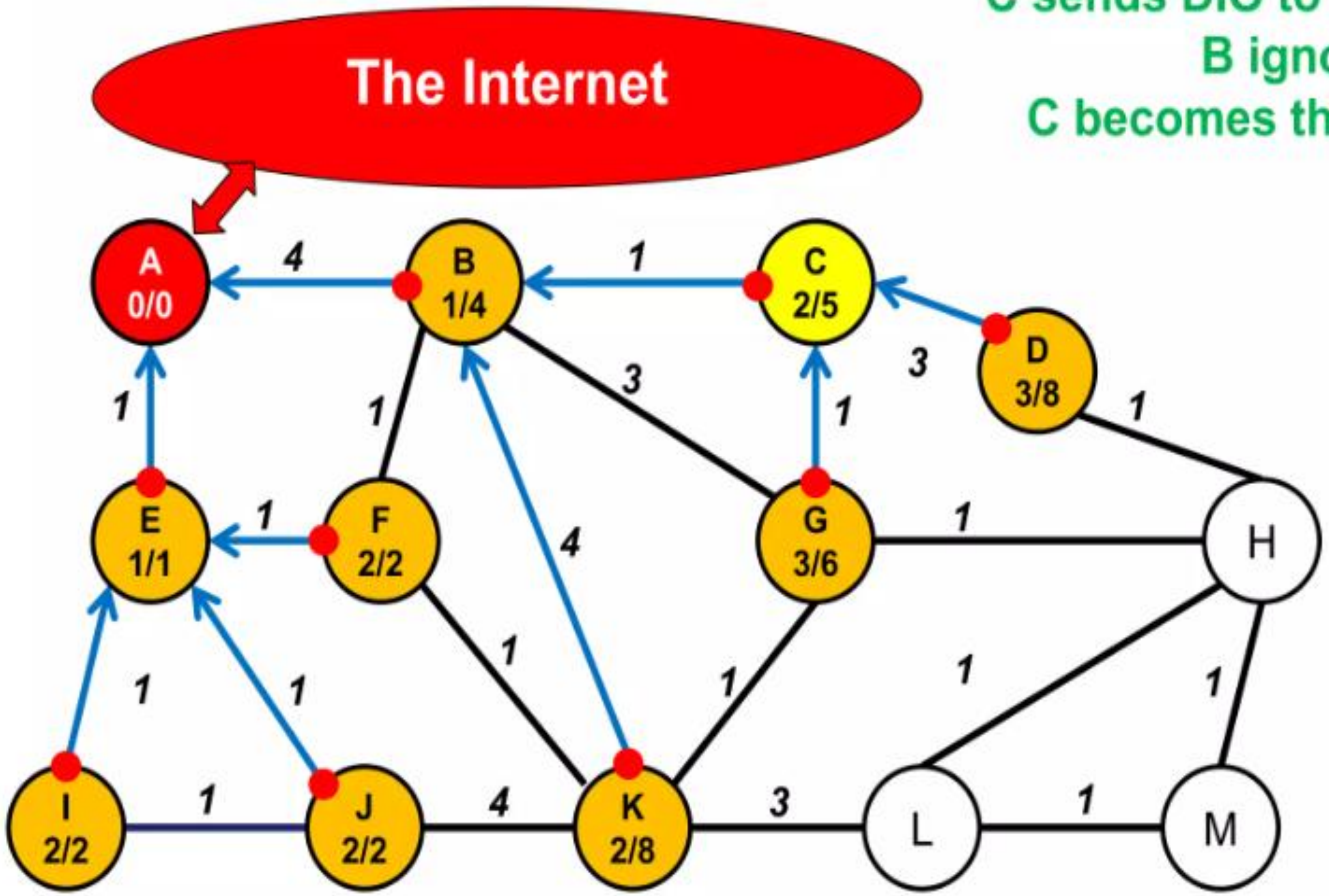


B sends DIO to A,C,F,G and K.
A ignores (rank).
B becomes the parent of all others.

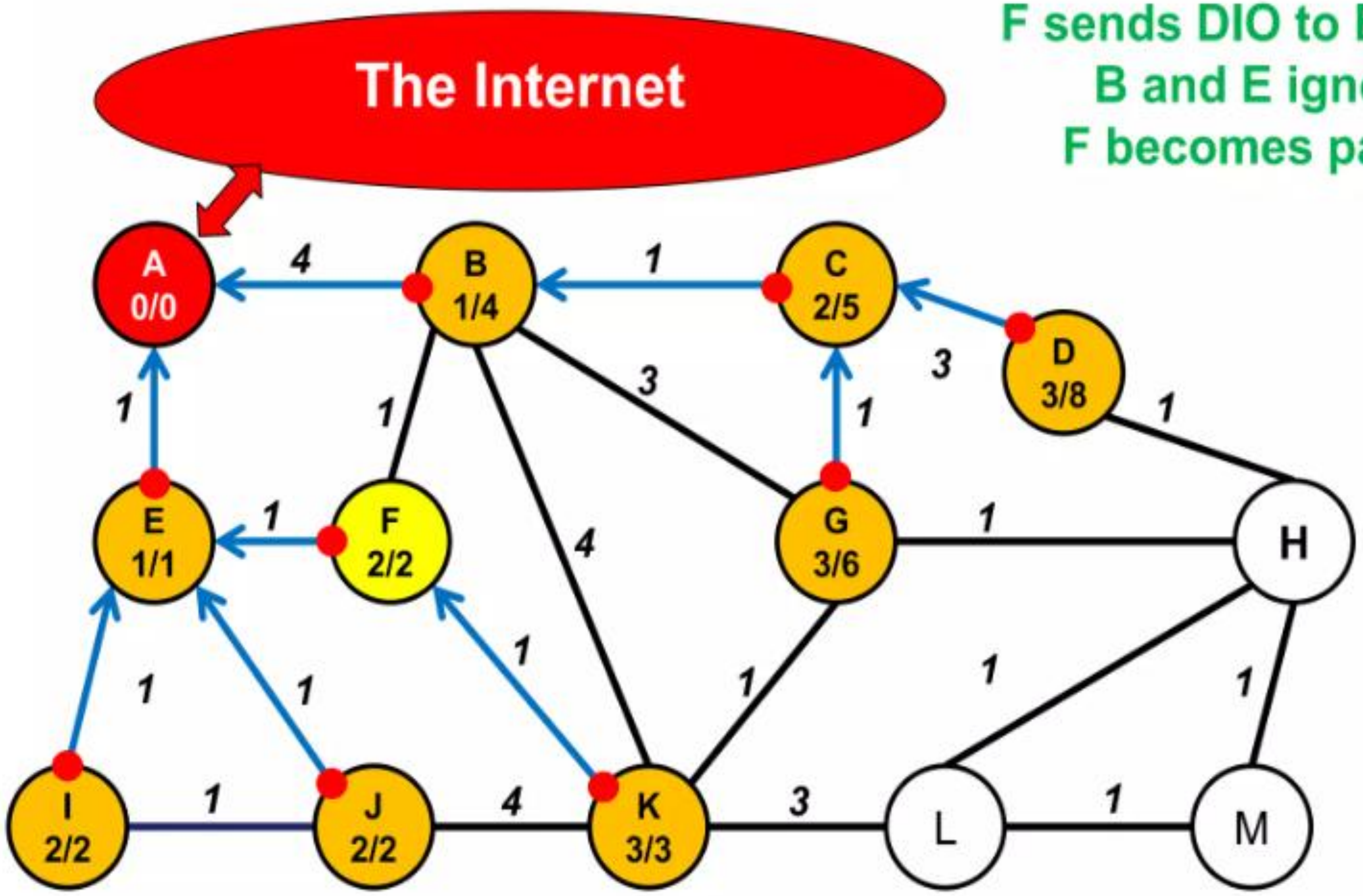
E sends DIO to A,F,I and J.
A ignores (rank).
E becomes the parent of F,I and J.

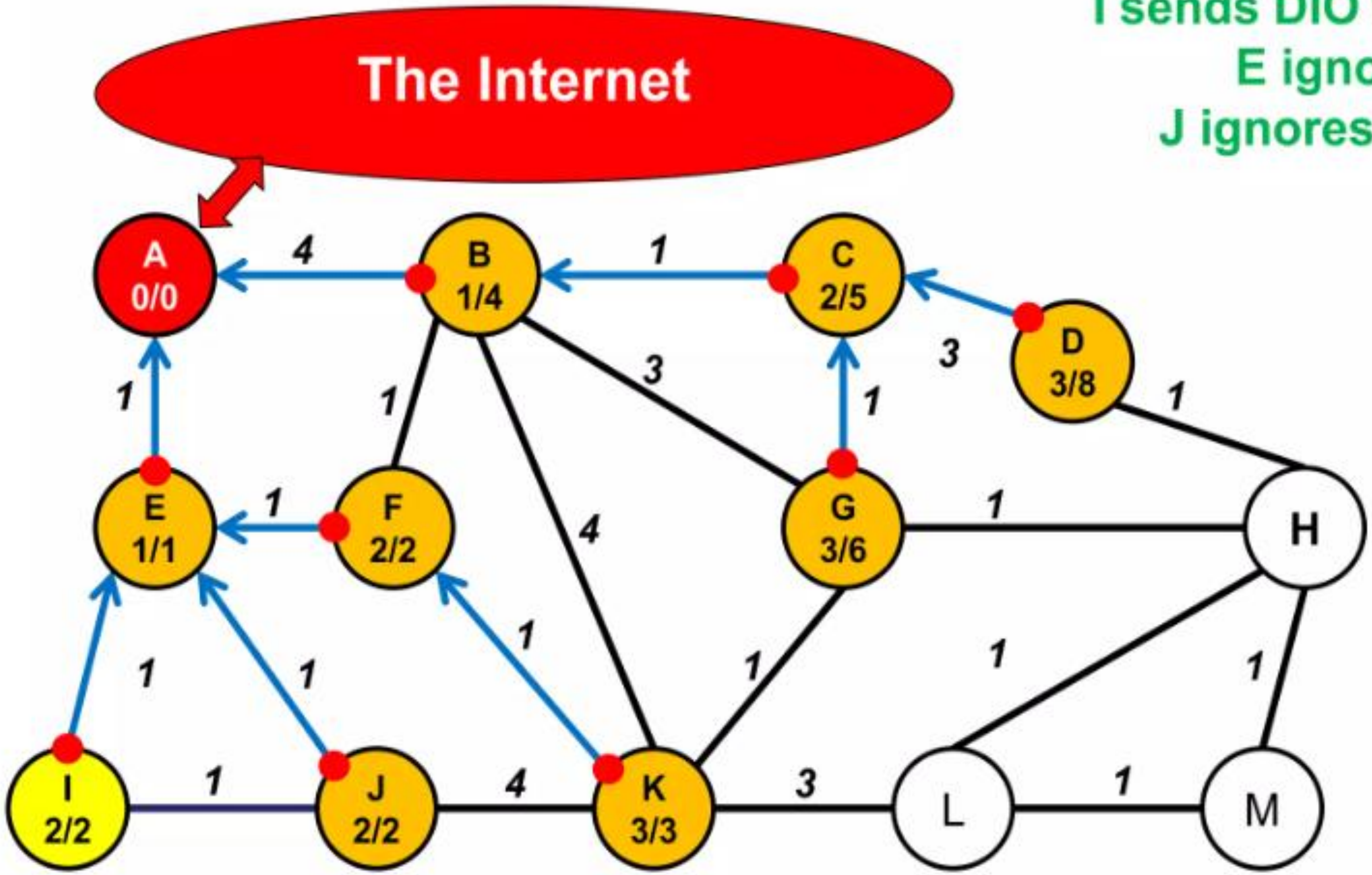


C sends DIO to B,D and G.
B ignores (rank).
C becomes the parent of
D and G.



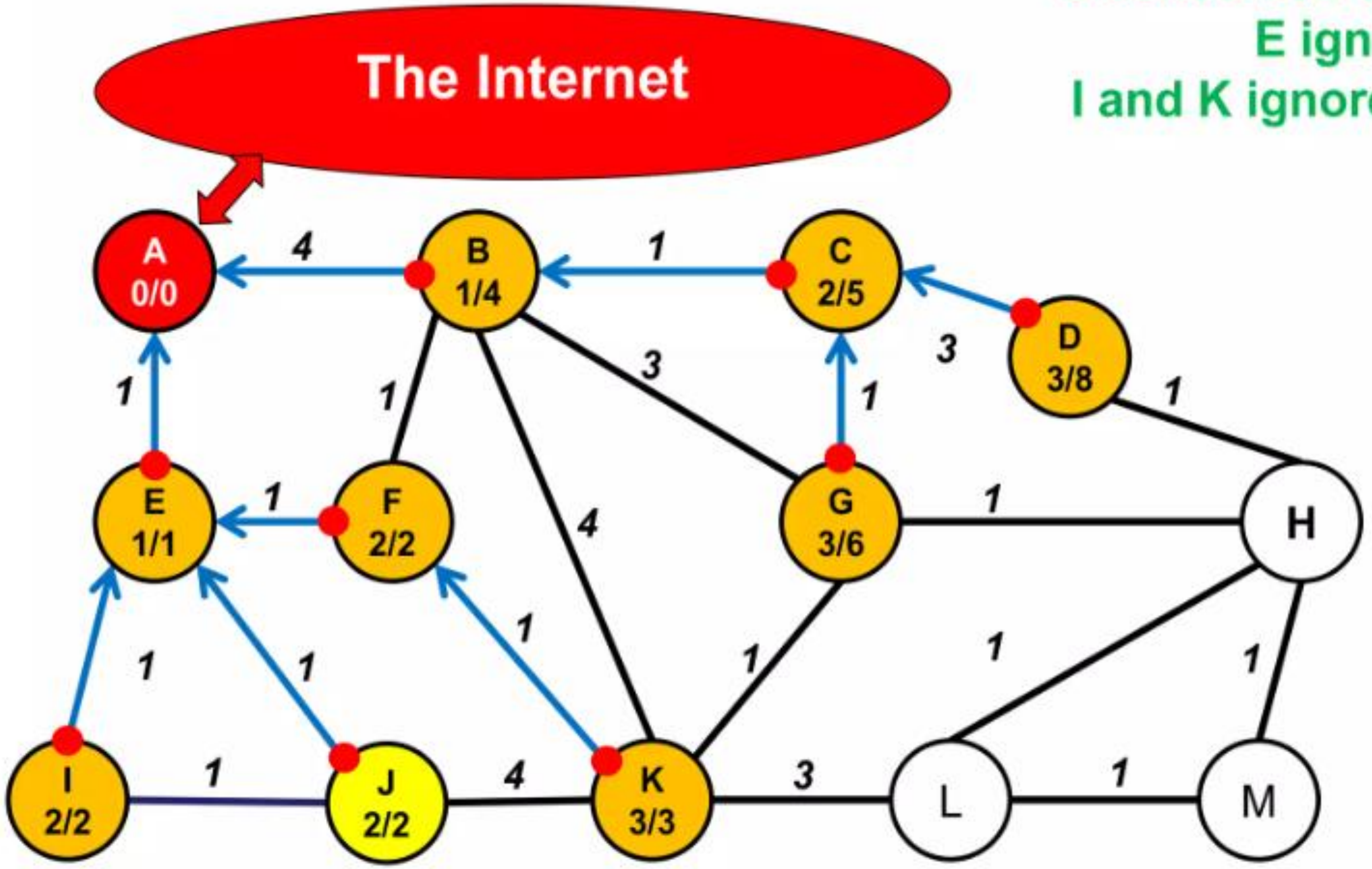
F sends DIO to B,E and K.
B and E ignore (rank),
F becomes parent of K.

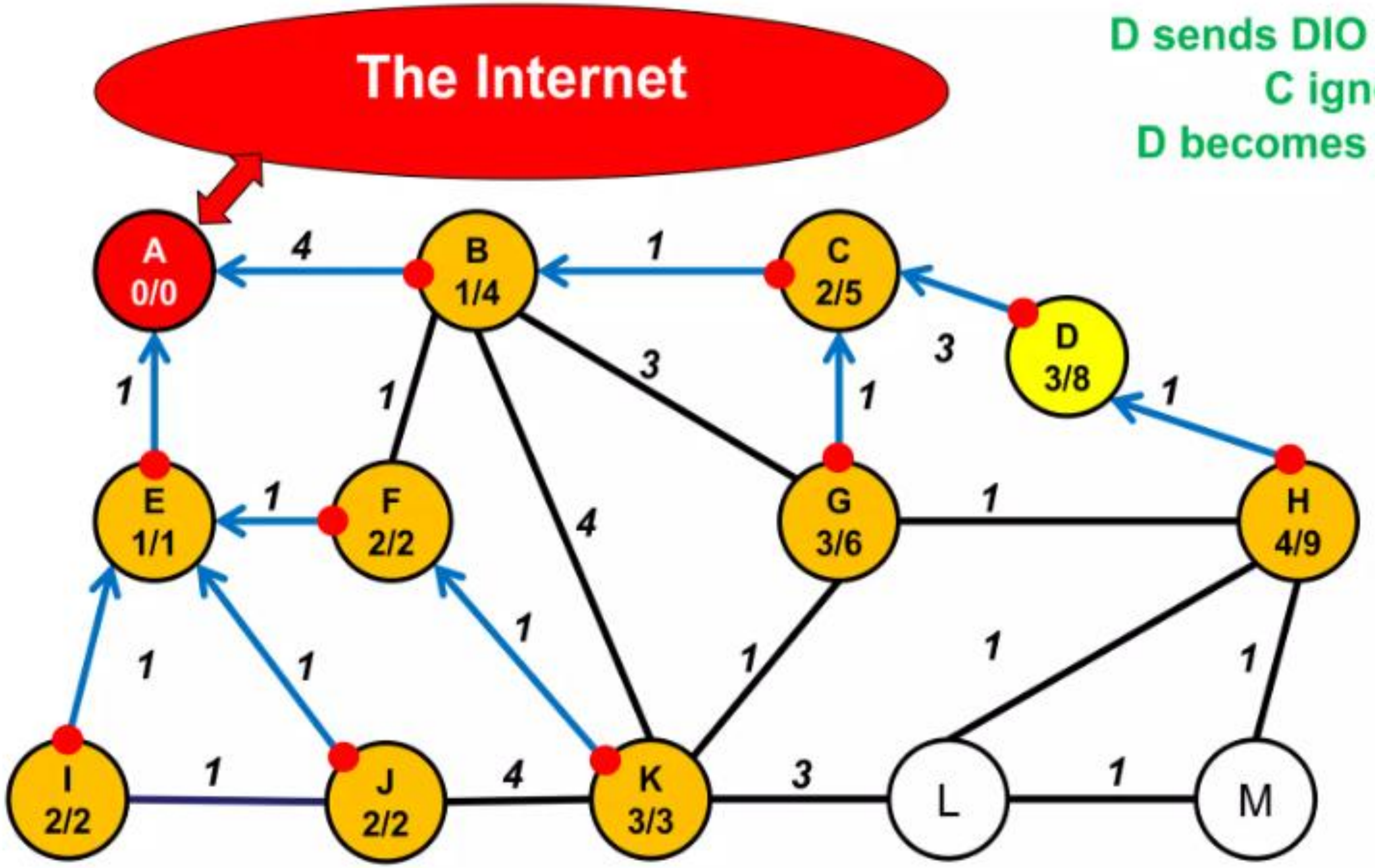




I sends DIO to E and J.
E ignores (rank),
J ignores (distance)

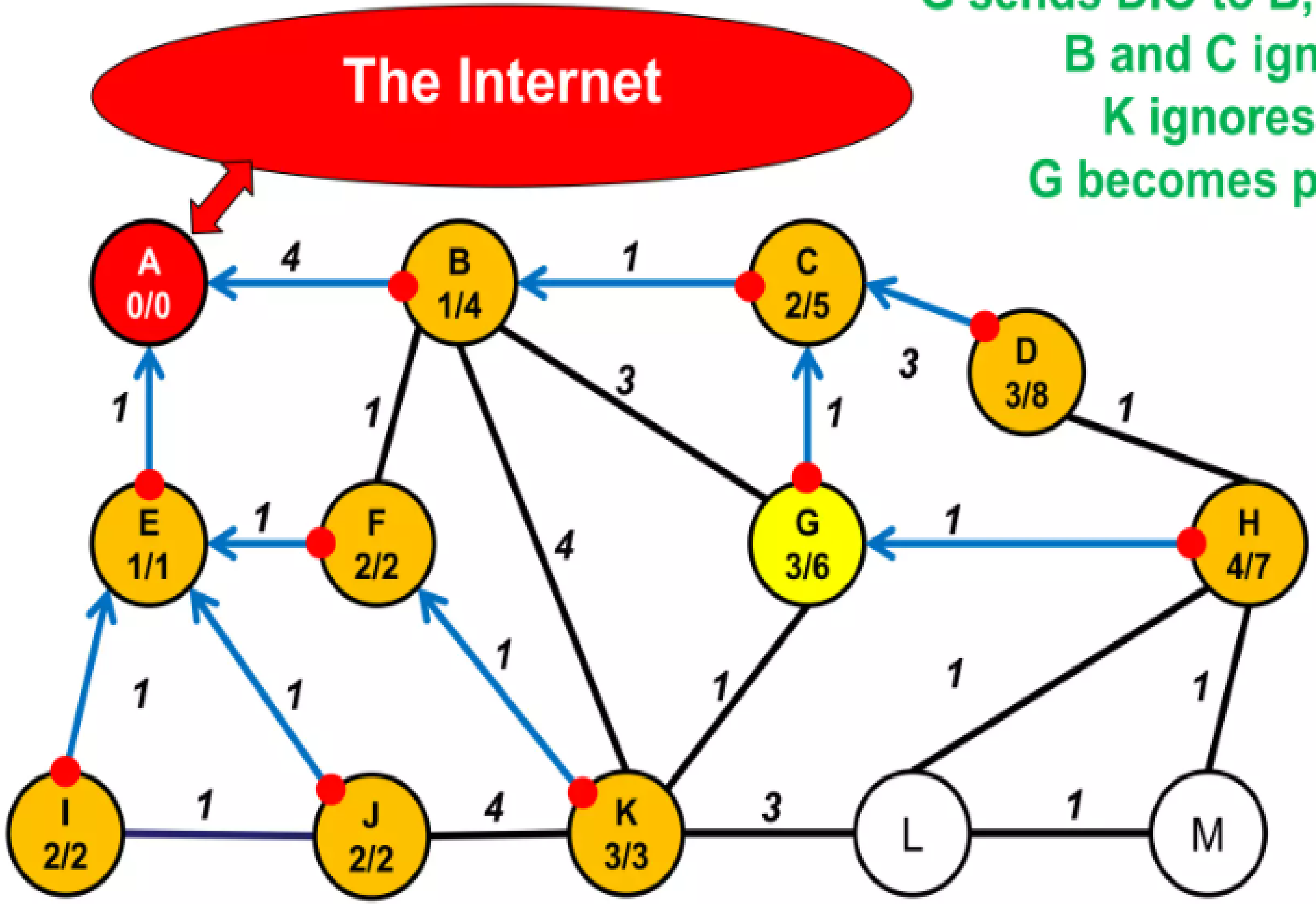
J sends DIO to E,I and K.
E ignores (rank),
I and K ignore (distance)



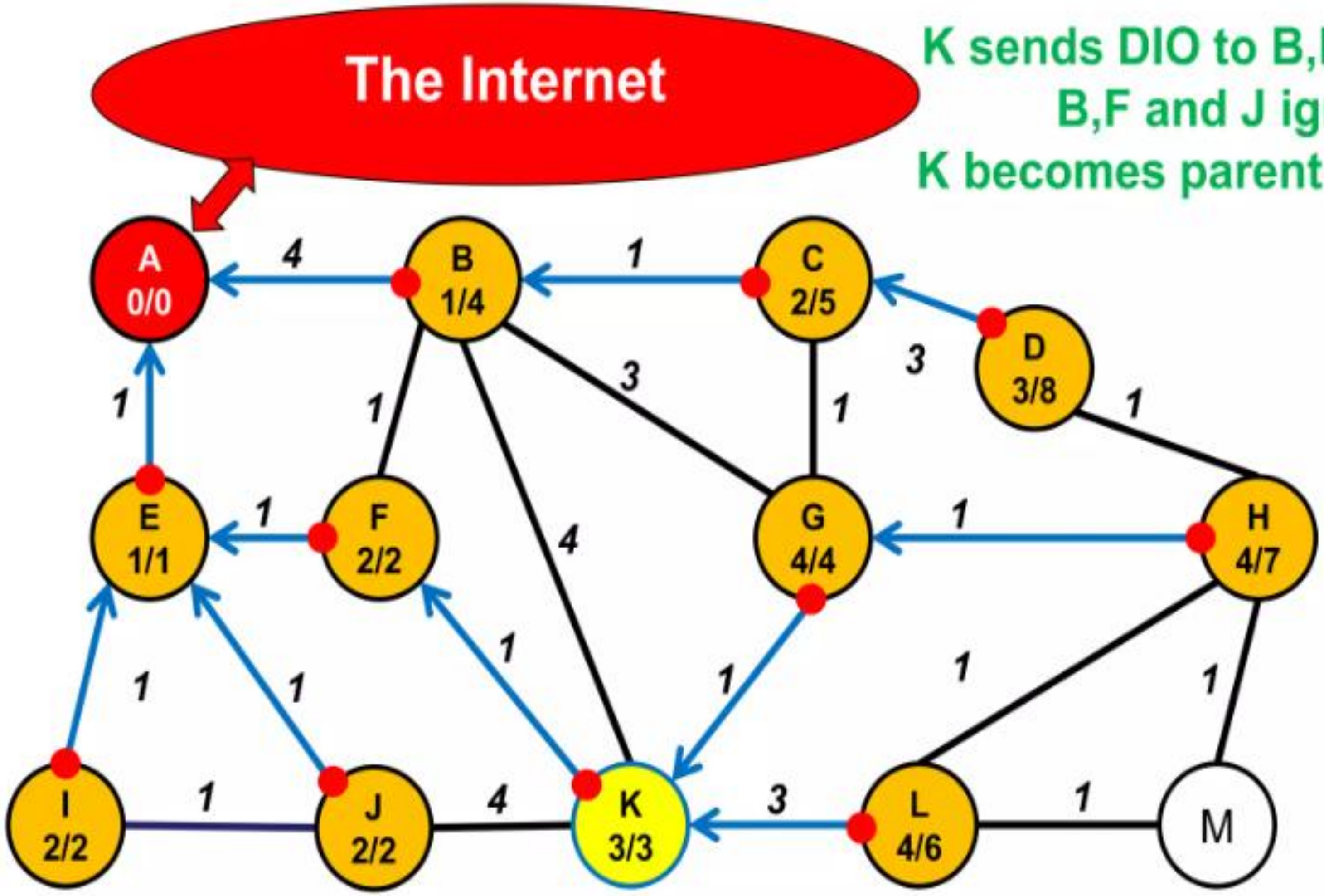


D sends DIO to C and H.
C ignores (rank),
D becomes parent of H

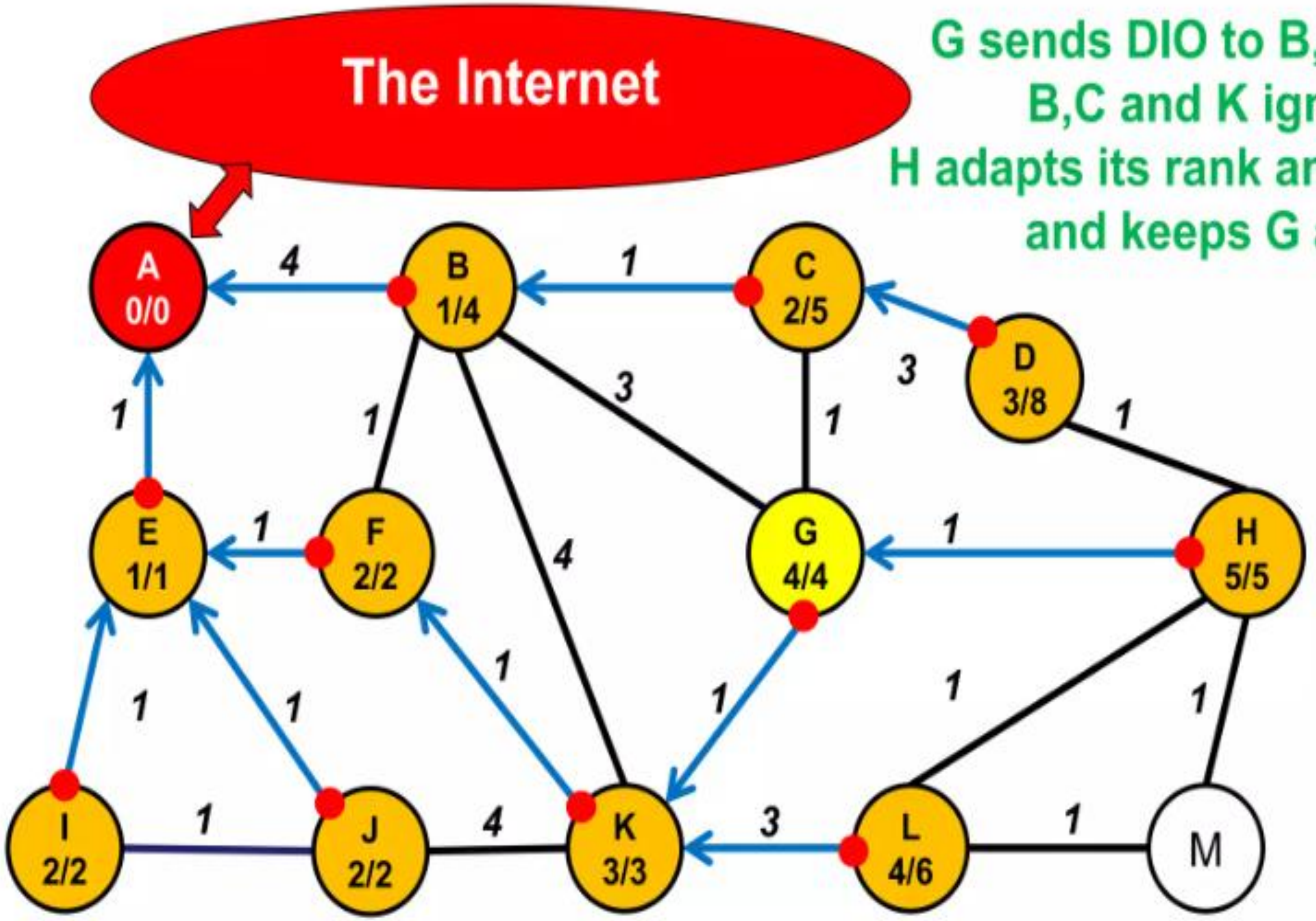
G sends DIO to B,C,H and K.
B and C ignore (rank),
K ignores (distance)
G becomes parent of H.

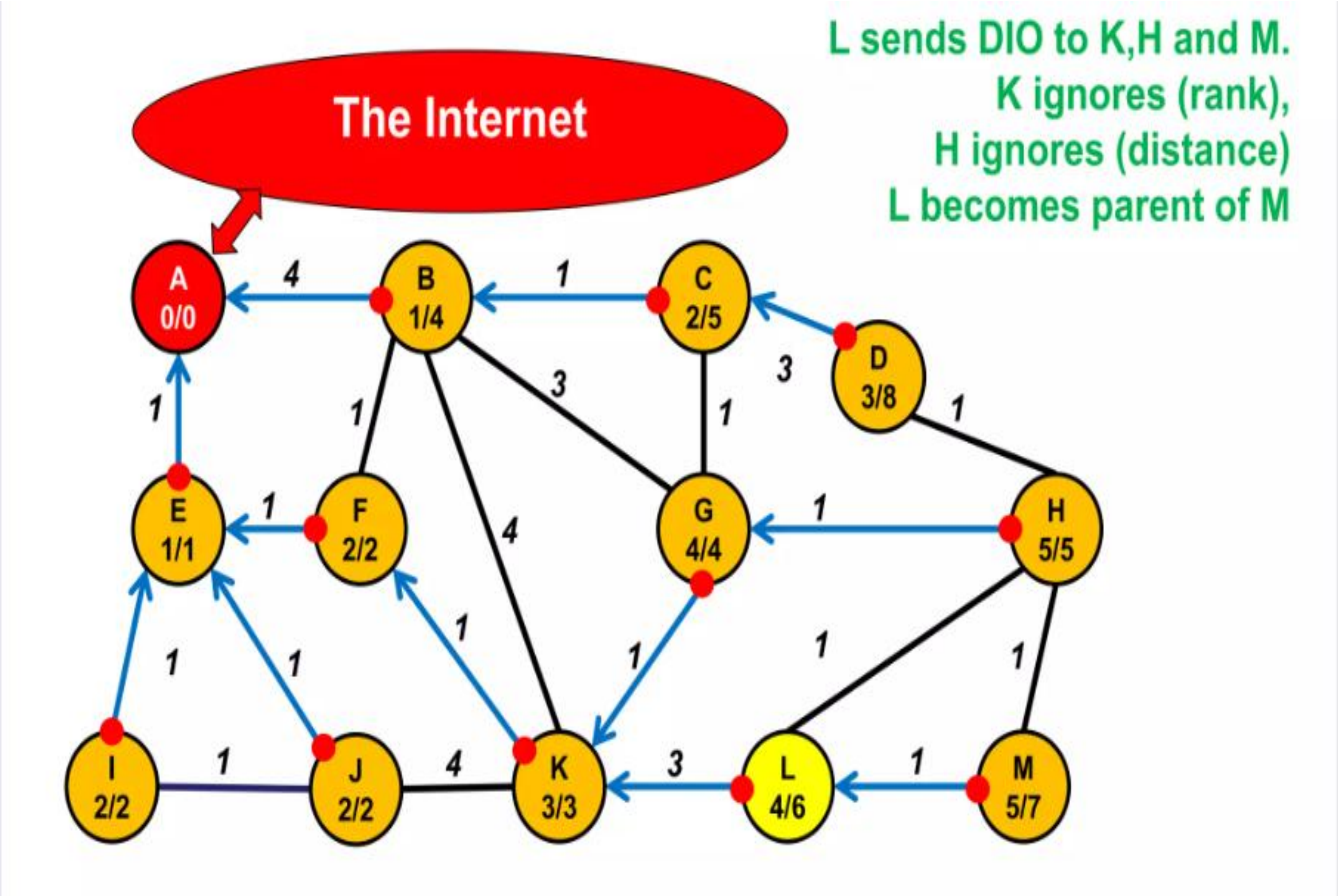


K sends DIO to B,F,G,J and L.
B,F and J ignore (rank),
K becomes parent of G and L.



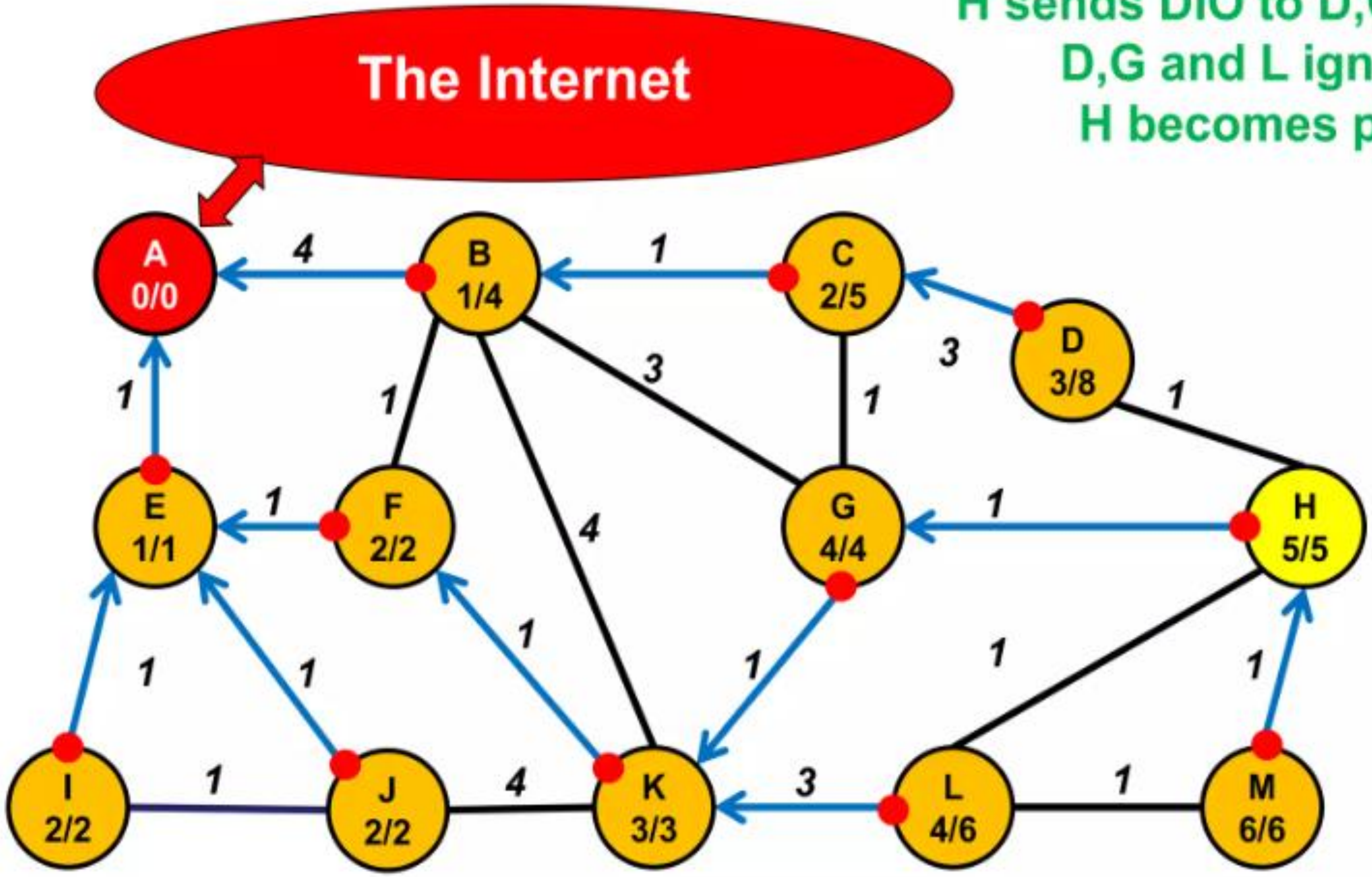
G sends DIO to B,C,H and K.
B,C and K ignore (rank),
H adapts its rank and distance
and keeps G as a parent



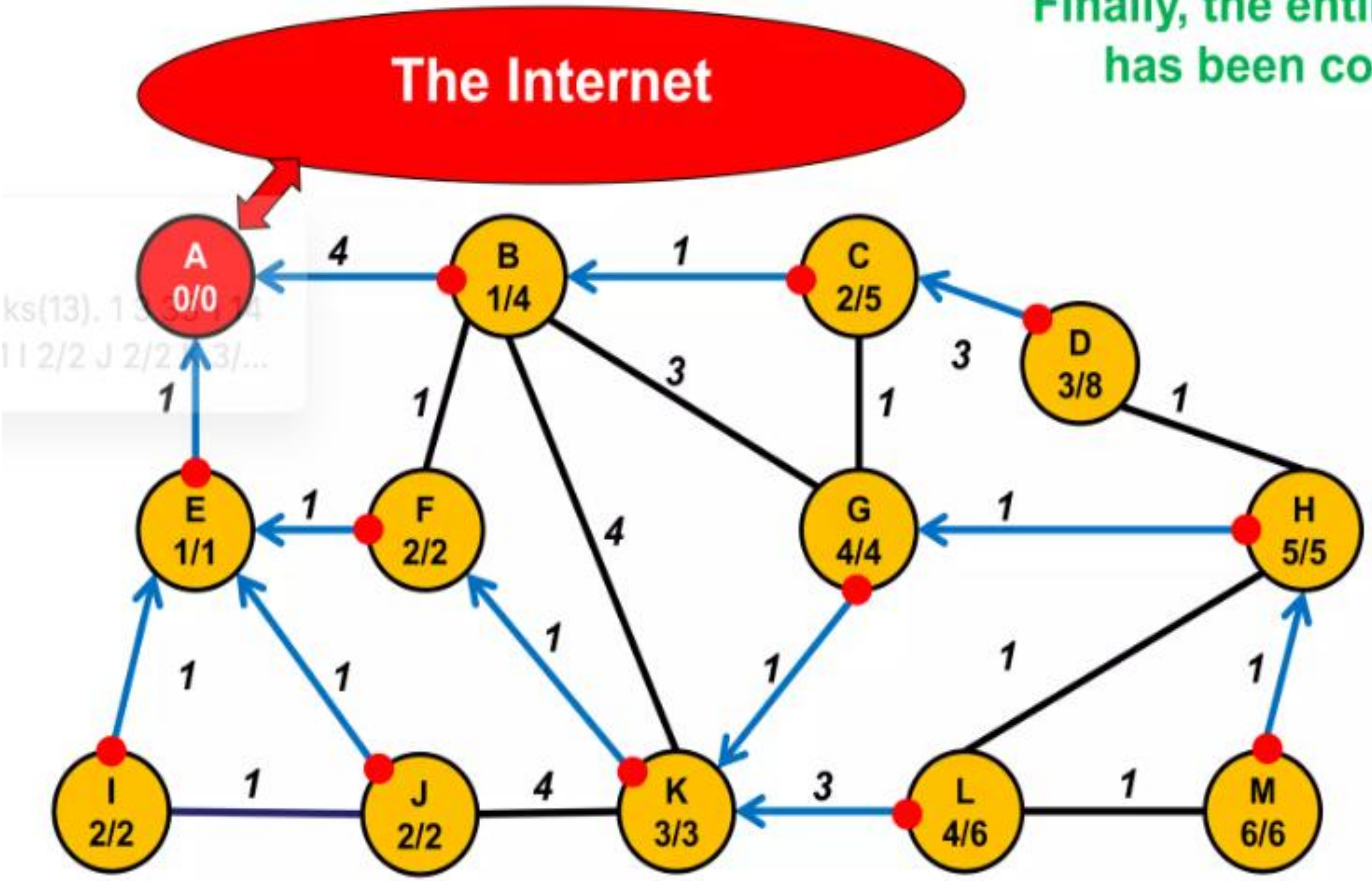


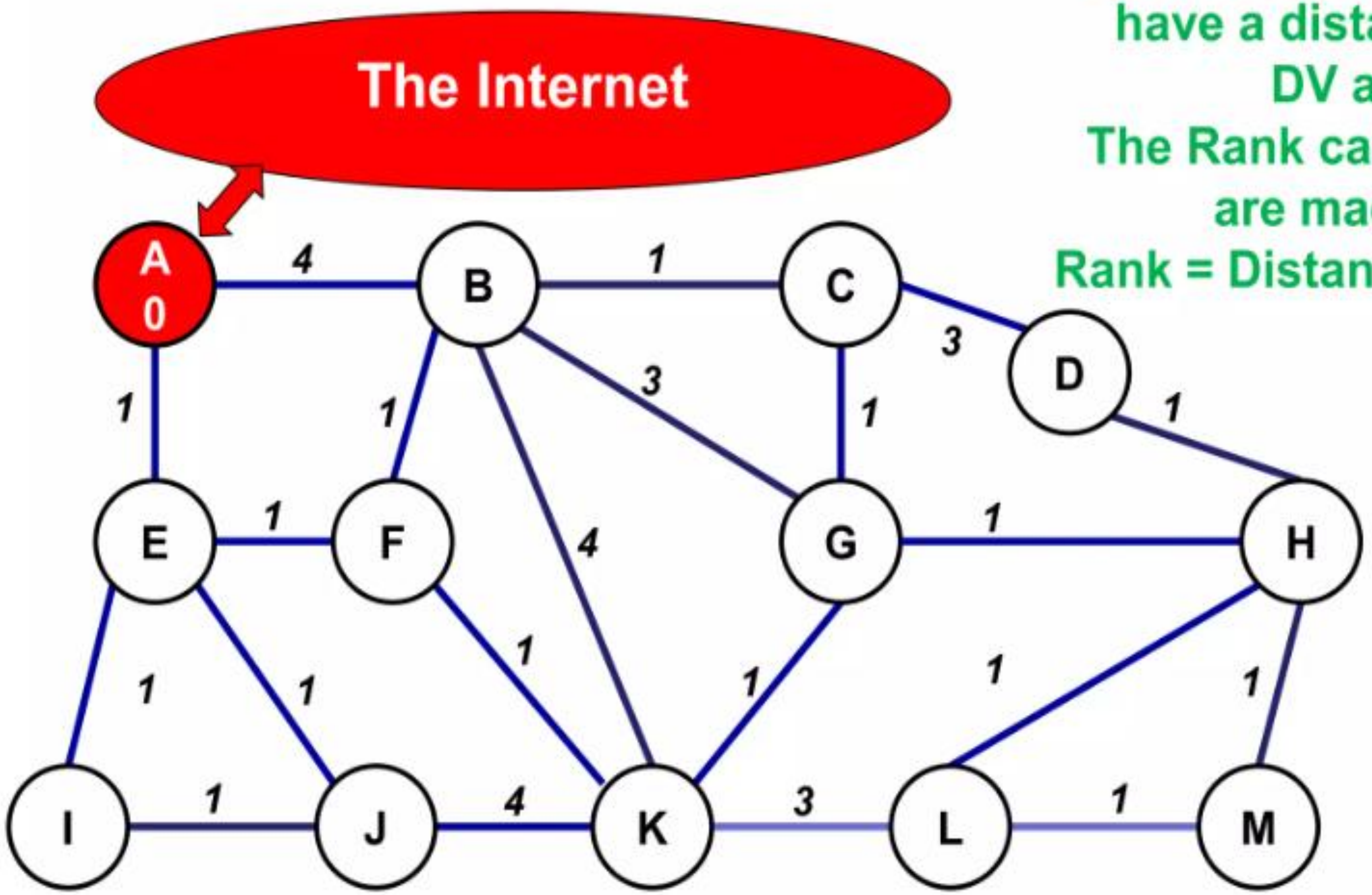
L sends DIO to K,H and M.
K ignores (rank),
H ignores (distance)
L becomes parent of M
The Internet
A
0/0
B
1/4
C
2/5
D
3/8
E
1/1
F
2/2
G
4/4
H
5/5
I
2/2
J
2/2
K
3/3
L
4/6
M
5/7
4
1
3
3
1
1
1
1
1
4
1
1
1
1
1
1
1
1
4
3
1

H sends DIO to D,G,L and M.
D,G and L ignore (rank),
H becomes parent of M

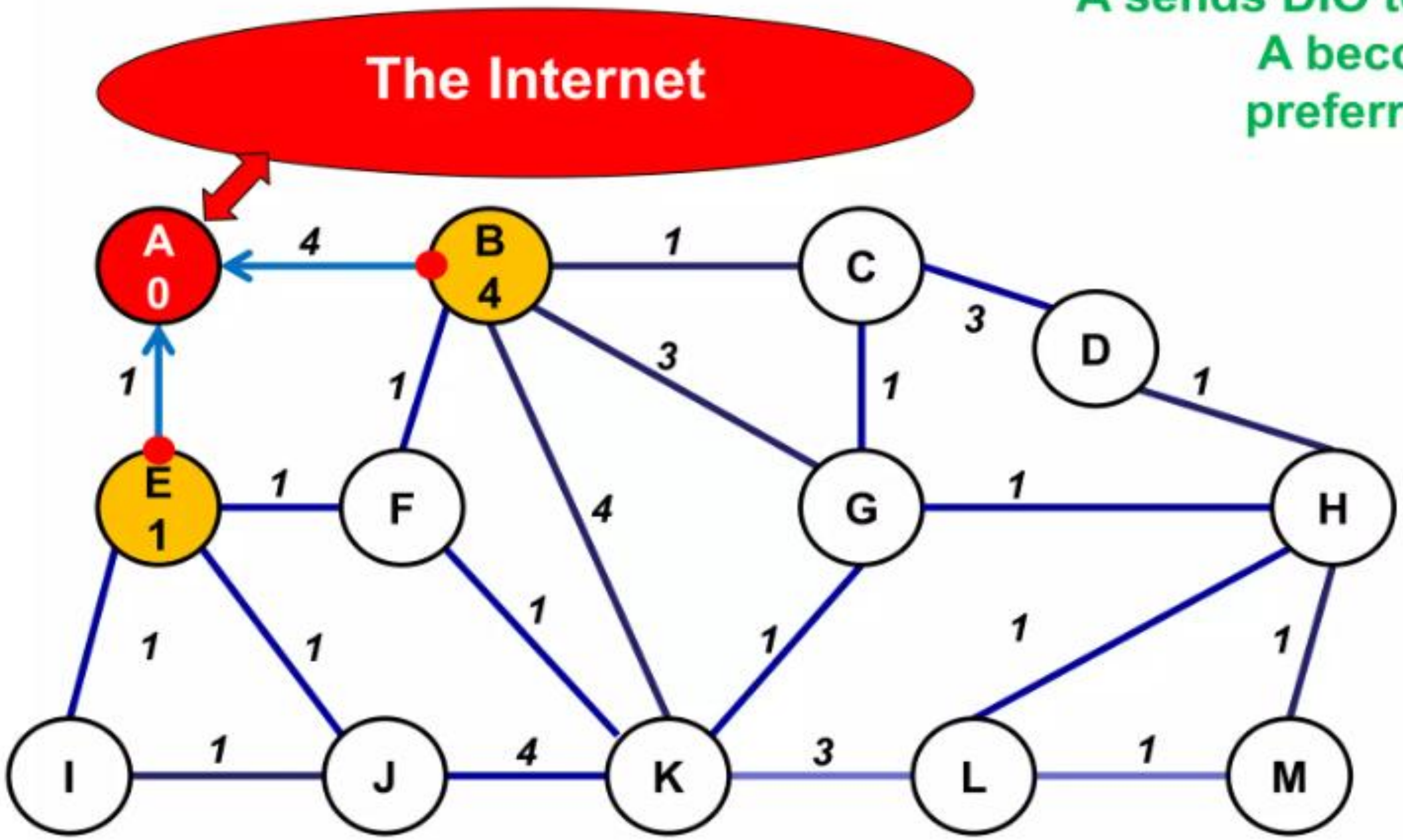


Finally, the entire DODAG has been constructed.



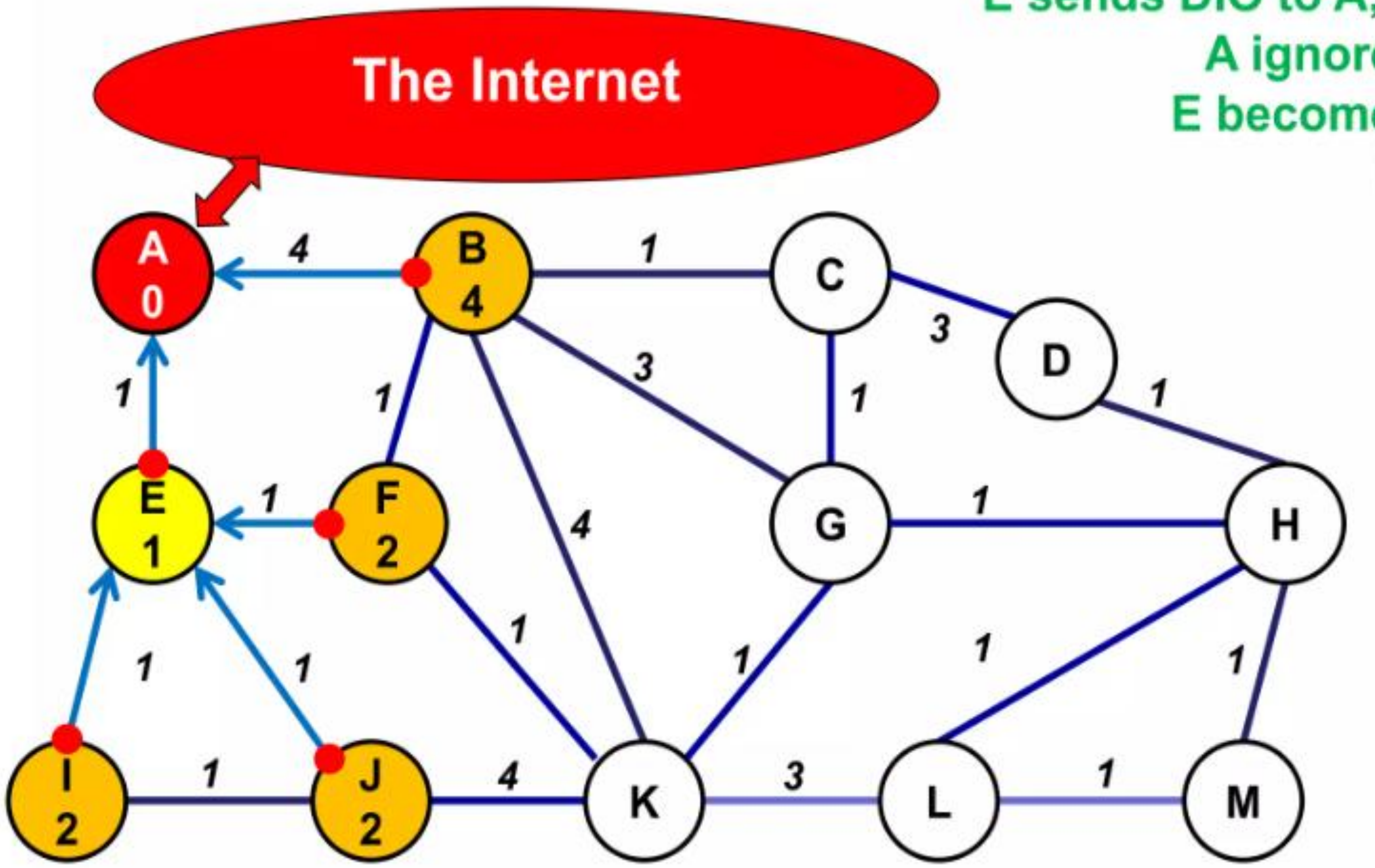


For this example, links have a distance as in DV and AODV.
The Rank calculations are made so that Rank = Distance to sink

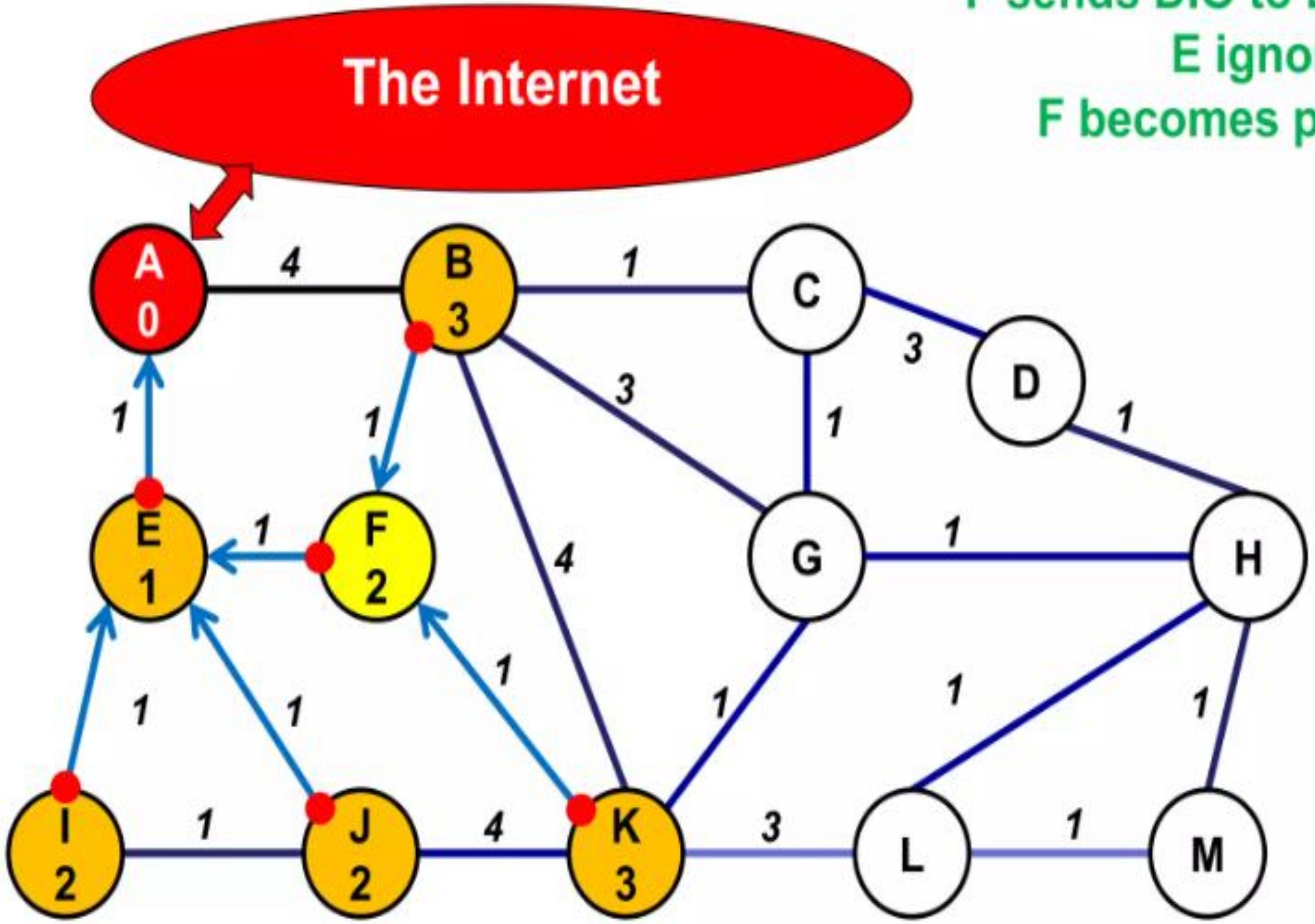


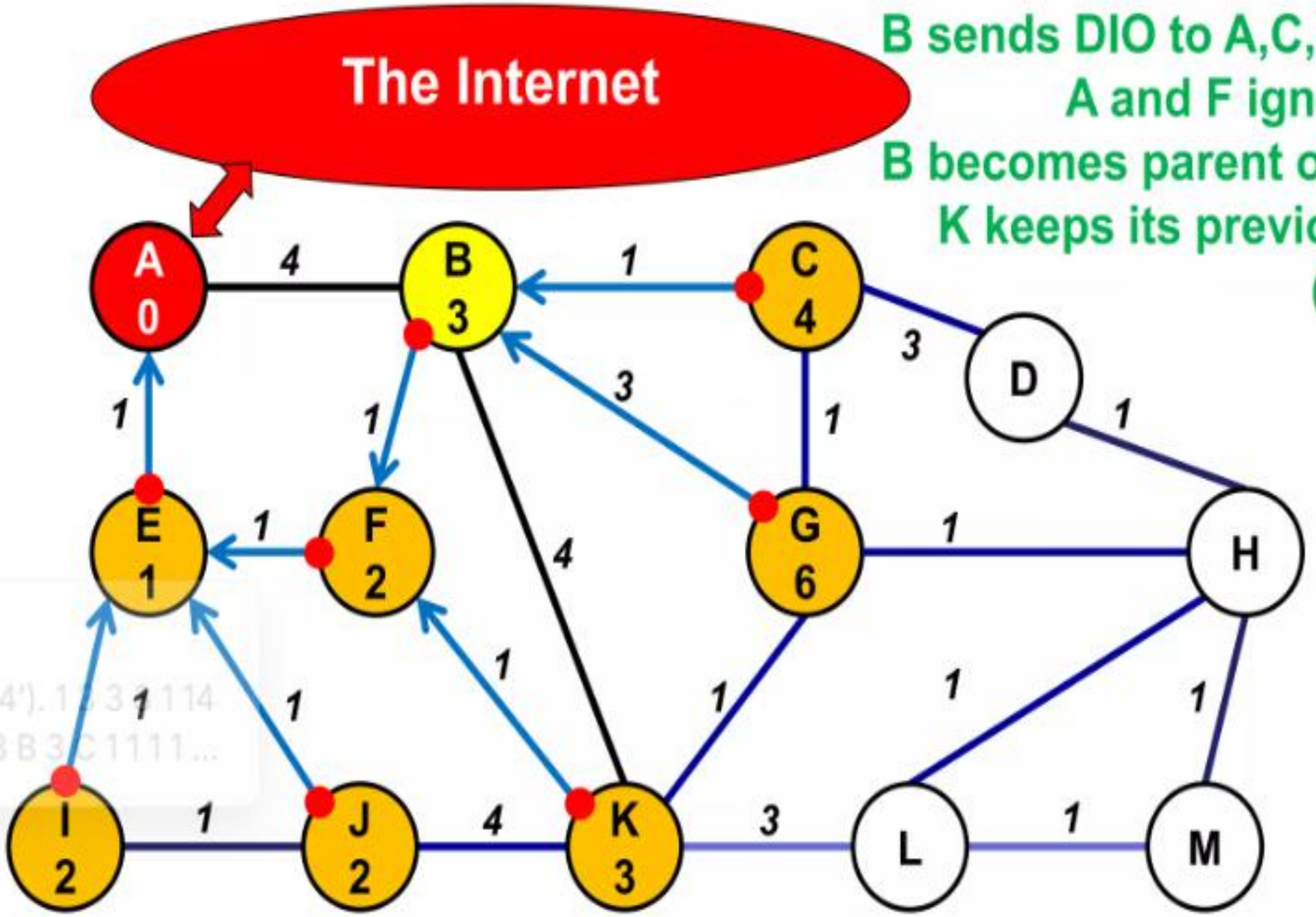
A sends DIO to B and E.
A becomes their
preferred parent.

E sends DIO to A,F,I and J.
A ignores (rank).
E becomes parent
F,I and J.

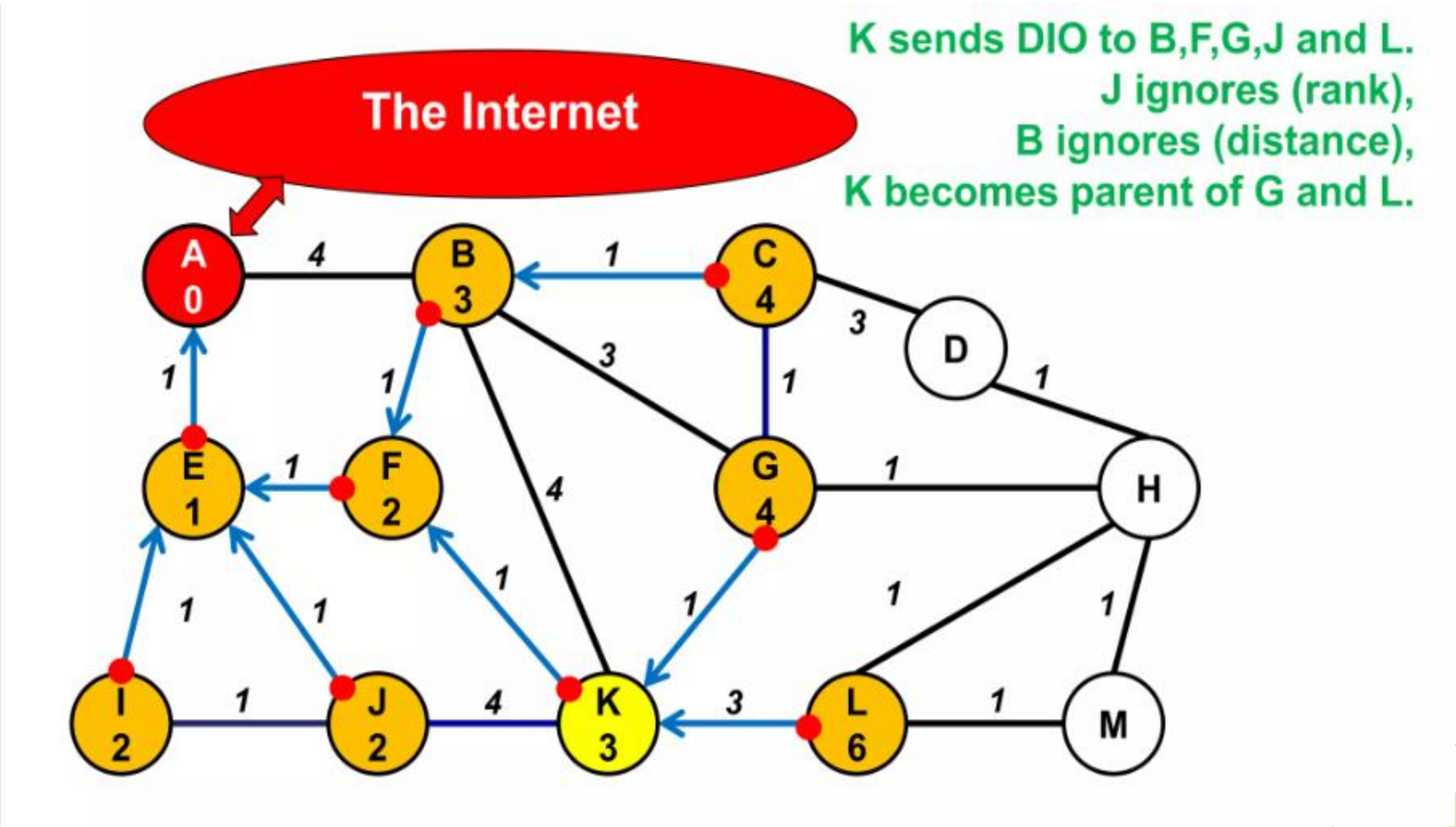


F sends DIO to B,E and K.
E ignores (rank).
F becomes parent of B
and K.

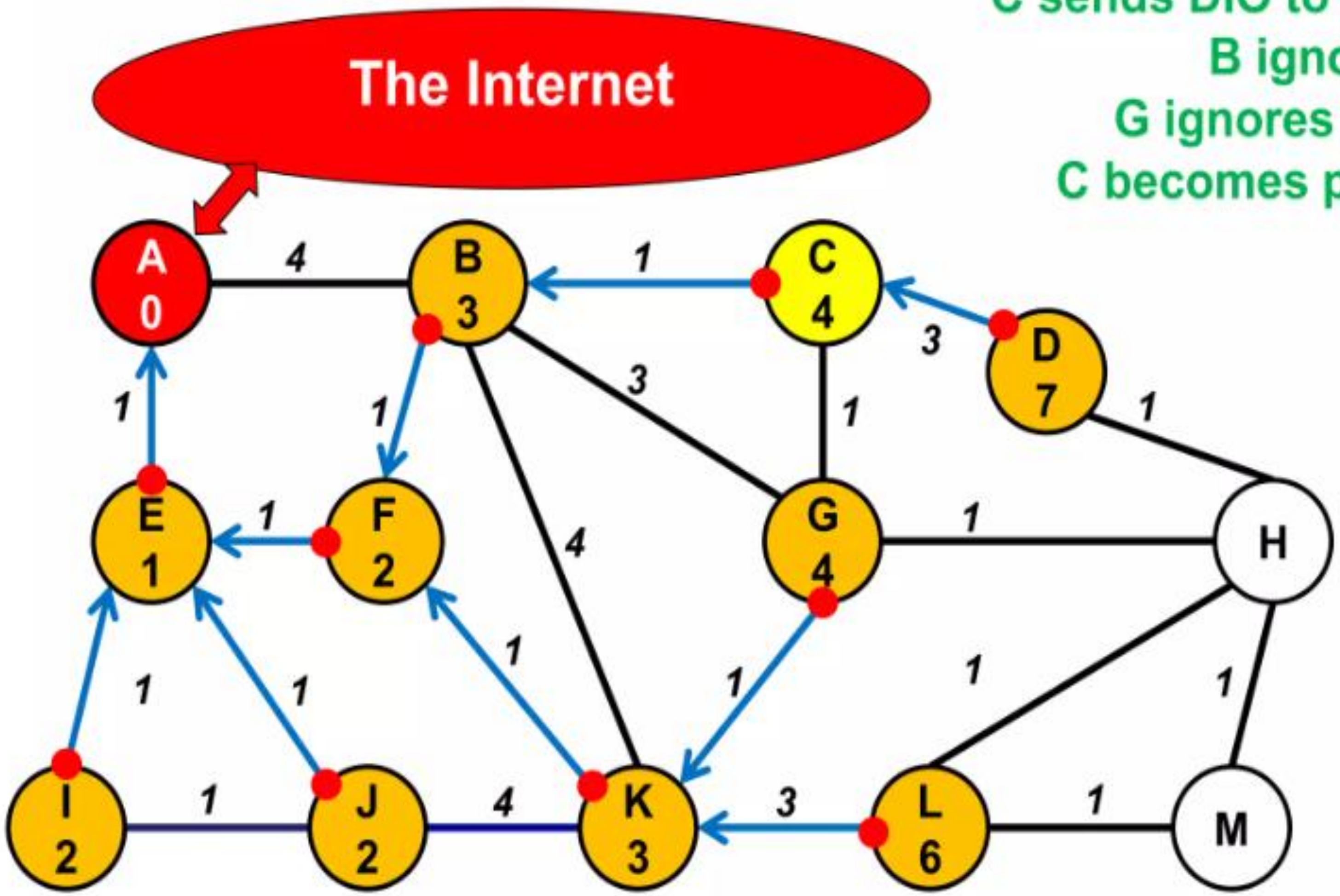




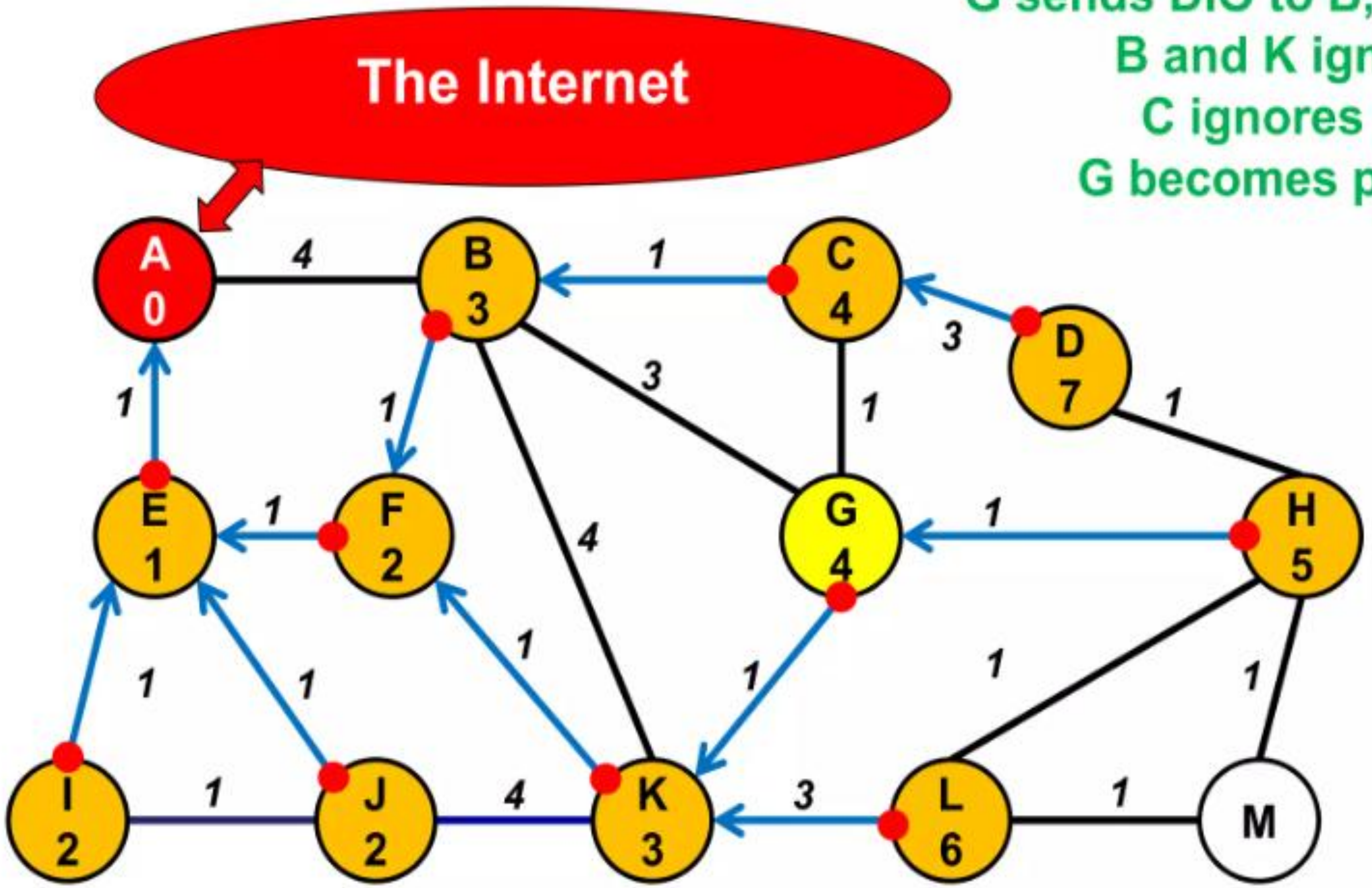
B sends DIO to A,C,F,G and K.
A and F ignore (rank).
B becomes parent of C and G.
K keeps its previous parent
(distance).


K sends DIO to B,F,G,J and L.
J ignores (rank),
B ignores (distance),
K becomes parent of G and L.
The Internet
A 0
B 3
C 4
D
E 1
F 2
G 4
H
I 2
J 2
K 3
L 6
M
4
1
3
3
1
1
1
1
1
4
1
1
1
1
1
1
1
4
3
1

C sends DIO to B,D and G.
B ignores (rank),
G ignores (distance),
C becomes parent of D.



G sends DIO to B,C,H and K.
B and K ignore (rank),
C ignores (distance),
G becomes parent of H.



H sends DIO to D,G,L and M.
G ignores (rank),
L ignores (distance),
H becomes parent of D and M.